

Factors influencing STEM participation and effective intervention strategies

ENESET Analytical report



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ABOUT ENESET

ENESET is an advisory network of experts specialising in the social and economic dimensions of education and training. It continues the work of previous networks coordinated for the European Commission's Directorate-General for Education and Culture. ENESET draws on the legacy of NESET and EENEE to support evidence-based policymaking through high-quality research, analysis, and policy advice. The network is coordinated by the Public Policy and Management Institute (PPMI). For inquiries, please contact us at: eneset@ppmi.lt.

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Factors influencing STEM participation and effective intervention strategies

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Executive summary

The goals of the European Union (EU) with regard to innovation, competitiveness and addressing major societal challenges such as the green and digital transitions significantly rely on strengthening the Union's science, technology, engineering and mathematics (STEM) workforce. Yet the EU faces persistent shortages of talent and underrepresentation within these fields – highlighting the challenges to be faced in achieving the ambitious target of increasing STEM's share in higher education enrolment from its 2022 level of 27.1 % to 32 % by 2030. The present report provides evidence-based insights to guide effective policy towards this goal. These insights are derived by synthesising international and EU-specific research on the factors that influence student choices regarding STEM pathways, and by evaluating the effectiveness of interventions designed to boost participation in STEM. Following a rigorous methodology, the analysis prioritises findings that are supported by causal evidence (from randomised controlled trials and quasi-experiments), drawing on research from both within and outside the Union. Given the diversity of the EU, all findings are assessed critically for their relevance and transferability across different national contexts.

Evidence shows that students' decisions are shaped by a multitude of factors operating cumulatively over the course of their development. Early influences – including supportive family environments, parental beliefs about the utility of STEM subjects, and engaging, high-quality experiences during school years – lay critical groundwork. As students progress through school, psychological and social factors become increasingly salient. Critical drivers include strong self-efficacy; intrinsic motivation linked to relevance or societal goals; the development of an identity as a “STEM Person” and an affinity for the subject matter – alongside the perception that one can “belong” and find a place within future STEM professions. Conversely, negative perceptions of STEM, stereotypes and anxiety about mathematics can deter potential students, while an individual's preference for (and willingness to choose) competitive academic and career environments, play a nuanced role in educational sorting. Furthermore, exposure to relatable role models as well as perceptions about the nature of post-secondary learning environments have a demonstrable impact on students' choices and subsequent persistence.

Evaluating interventions designed to address these factors shows that different approaches exhibit varying levels of effectiveness. Interventions based on strong non-EU causal evidence and which have lower estimated costs, such as informing parents about the utility of STEM subjects or framing STEM in social terms, warrant rigorous testing within the EU. Brief exposure to professional role models emerges as a high priority. This is supported by strong EU causal evidence and low-to-medium costs. Other interventions demonstrate causal effects within the EU but entail higher costs. These include the early use of digital tools to boost girls' confidence; hands-on, creative activities (e.g. FabLabs); and structured study counselling. EU-based evidence suggests that integrated

curricula also fall into this “proven but costly” category. Some interventions that have proved to be effective outside the EU, such as learning communities, appear more resource-intensive and will require careful consideration in their adaptation.

Crucially, however, such targeted interventions alone are insufficient. This report concludes that systemic reforms are an essential prerequisite in order to substantially increase participation in STEM. Individuals’ choices and the success of interventions are fundamentally shaped by the structure and resourcing of the education system concerned. Specifically, enhancing the permeability and effectiveness of pathways from vocational education and training (VET) into STEM higher education offers a strategic opportunity to significantly expand the talent pool. Simultaneously, higher education funding models in Member States must properly recognise and account for the higher per-student costs of quality STEM education, to avoid creating disincentives for institutions to expand their STEM provision, which could lead to limited enrolment capacity and threaten quality.

Despite having identified the critical factors above – as well as potentially effective strategies – significant gaps hinder optimal policy design. There is a pressing need for more rigorous, EU-based causal evaluations of many interventions; in particular, testing the transferability of promising interventions or approaches identified outside the EU. Furthermore, a pervasive lack of comprehensive, comparable data on cost-effectiveness limits efficient prioritisation. The causal impacts of specific reforms to funding and VET pathways also require dedicated research, alongside EU-focused longitudinal studies to understand student trajectories.

Based on this analysis, the key recommendations for the EU and Member States provided in this report focus on several priorities. First, it is crucial to support high-quality evaluations (causal, cost-benefit) of promising interventions within diverse EU contexts. Second, it is vital to promote policy learning and knowledge exchange about systemic reforms – in particular, cost-aware higher education (HE) funding and effective VET-to-HE pathways, supported by better data. Third, consistent investment is needed in interventions that address the stereotypes and negative perceptions of STEM that students encounter from an early age. Lastly, encouragement should be given to the identification, evaluation and dissemination of potentially cost-effective approaches, such as information campaigns and well-designed role model programmes.

1. Introduction

Science, technology, engineering and mathematics (STEM) are increasingly recognised as critical drivers of economic competitiveness, innovation and sustainable growth in the European Union (EU). A robust STEM workforce is essential in order to address key societal challenges including labour shortages, skill gaps, achieving the goals of the European Green Deal and facilitating the digital transition (European Commission, 2020). Investments in STEM education and careers are, therefore, paramount to securing the EU's future prosperity and global leadership in the 21st century.

Despite the acknowledged importance of STEM, the EU faces persistent challenges in relation to an insufficient supply of STEM talent (Deming & Noray, 2018; Kahn & Ginther, 2017). This issue was recognised as a central concern for the EU in the Commission's recent Communication on the "Union of Skills", which frames the mitigation of skills shortages – especially those in STEM – as fundamental to increasing the EU's competitiveness and achieving its strategic goals (European Commission, 2025). The shortage of talent presents a significant challenge to meeting the European Commission's ambitious goal, proposed in the in its Union of Skills initiative and in the STEM Education Strategic Plan, to increase the share of enrolment in STEM subjects in higher education from 27.1 % in 2022 to 32 % by 2030. The current shortage, coupled with the significant underrepresentation of certain demographic groups, especially women, within key STEM fields such as engineering, information technology and physics (Cerinsek et al., 2013; Kahn & Ginther, 2017), negatively impacts the EU's capacity to innovate, compete globally and achieve its strategic goals.

Understanding why students either choose or reject STEM pathways is a complex task. Such choices are influenced by factors ranging from early life experiences (as noted by Harackiewicz et al., 2012; Starr et al., 2022; Wai et al., 2010); psychological variables such as self-efficacy, identity and competitiveness (see Buser et al., 2014; Holmegaard et al., 2014; Wang, 2013); post-secondary environments including teaching and mentoring (see Carrell et al., 2009; Coppens et al., 2023; Mansour et al., 2020); and broader career considerations (see Cerinsek et al., 2013; Deming & Noray, 2020).

1.1. The present report

This report aims to provide evidence-based insights to guide EU policies aimed at increasing participation in STEM. The report draws upon a wide range of international research while prioritising findings that originate from (or are relevant to) the EU context. To that end, this investigation will first seek to identify the pivotal factors that influence decisions to pursue STEM education, across individual, social and institutional levels. Crucially, it will critically evaluate the

quality of evidence supporting the influence of these factors, prioritising insights from studies that employ rigorous methods capable of inferring causality, such as randomised controlled trials (RCTs) (e.g. Ballerini et al., 2024; Beckmann & Fervers, 2024; Breda et al., 2021; Chirikov et al., 2020; Harackiewicz et al., 2012; Porter & Serra, 2020) and quasi-experimental designs (e.g. Carrell et al., 2009; Means et al., 2016; Russell, 2017), while carefully considering findings from longitudinal and observational research (e.g. Bieri Buschor et al., 2014; Holmegaard et al., 2014; Wai et al., 2010; Wang, 2013). The report then turns to an evaluation of the effectiveness of diverse interventions designed to amplify STEM participation within higher education. These include educational programmes; curriculum reforms Anwar et al., 2022; De Meester et al., 2015; Knipprath et al., 2018); targeted support such as counselling or mentoring (Beckmann & Fervers, 2024; Dawson et al., 2015); role model initiatives (Besecke & Reilly, 2006; Breda et al., 2021; Porter & Serra, 2020); pedagogical approaches (Diekman et al., 2011; Thibaut et al., 2018); and the provision of information (Ding et al., 2021). Our assessment includes, where possible, considerations of the cost-effectiveness of these intervention designs. Lastly, following the accumulation of this body of knowledge, the report formulates direct and actionable recommendations intended to advance STEM education and to strengthen the development of the STEM workforce throughout the EU.

This report tackles two fundamental and interconnected research questions. The first of these is to identify which segments of the population are drawn towards STEM fields, and to explore the multifaceted motivations that underlie those choices (Wang & Degol, 2013) – with a particular focus on revealing the variations and nuances that exist between EU Member States and how these relate to the achievement of EU STEM targets. Second, we identify which intervention designs have been examined through rigorous and clear scientific research (specifically, with regard to causal impact) in order to make robust recommendations that can positively improve the quantity, quality and diversity of STEM graduates and researchers within the EU. Here, we discuss the potential benefits of attracting VET students towards STEM graduate programmes (Van den Broeck et al., 2018, 2019, 2020).

1.2. Systemic and institutional context of STEM participation in the EU

While this report focuses primarily on the factors that influence STEM choices at an individual level (in part due to the prevalence of research addressing such influences), it is crucial to acknowledge that these individual decisions are made within a complex system of systemic and institutional factors that vary significantly across the EU. These broader contexts shape opportunities, create constraints and ultimately influence the landscape within which individuals make choices about pursuing STEM education.

Educational systems in the EU are very diverse. Member States employ different approaches to curriculum design and teacher training (De Meester et al., 2021), as well as to the structure of secondary and tertiary education. Structural differences – such as early tracking systems (as seen in Germany, Austria) versus more comprehensive models (e.g. in Finland, Sweden) – directly shape the pool of students prepared for and eligible to choose STEM pathways in higher education (Hanushek & Woessmann, 2006). Similarly, variations in university admission policies – ranging from a reliance on national standardised tests (common in parts of Eastern Europe) or competitive entrance exams (for certain schools in France), to an emphasis on final secondary school grades (e.g. Germany’s Abitur, and common in countries such as Finland and Sweden) or diagnostic pre-enrolment tests (Hanssens et al., 2023; 2025; Pinxten et al., 2019b) – can influence the preparation of students, their expectations, and ultimately their access to STEM fields.

Funding models for higher education also differ markedly across the EU, impacting both student access and institutional capacity. Some Member States offer tuition-free higher education for EU students (e.g. Germany, the Nordic countries), with substantial public investment; others rely more heavily on student tuition fees (as in Ireland and the Netherlands). Furthermore, the mechanisms used to allocate core public funding to universities are highly varied, often combining elements such as input-based funding formulas linked to student numbers (common across many systems). Increasingly, such mechanisms incorporate output or performance metrics (e.g. as in Belgium, the Netherlands, and parts of the UK system that potentially influence EU practices), as well as negotiated performance contracts (e.g. in Finland, Austria), alongside historical allocations in various configurations across the Member States (Pruvot & Estermann, 2022).

The funding of STEM fields is especially complex, as these are often significantly more costly to deliver than disciplines such as humanities or social sciences. Studies and institutional analyses suggest that STEM disciplines can cost between 1.5 and 2.5 times more per student than baseline humanities subjects in many systems. This difference can be attributed to the specialised infrastructure, consumables, high-tech equipment requirements and typically lower student-staff ratios necessary for lab work and specialised instruction (see Table 1, derived from OECD comparisons; OECD, 2019).

Table 1. Subject-area weightings (cost factors) in selected OECD jurisdictions

	Flemish Community		Netherlands		Ireland	Denmark (Universities)	Finland	Portugal	
	Univ.	UC	Univ.	UAS				Univ.	Poly.

	Flemish Community		Netherlands		Ireland	Denmark (Universities)	Finland	Portugal	
Non-laboratory subjects (e.g. humanities and social sciences)	1	1	1	1	1	1	1	1.15 or 1.2	1
Subjects involving fieldwork (e.g. computer science, education)	2	1.1 to 1.6	1.5	1.28	1.3	1.4	1	1.6 to 1.9	1.3 to 2
Laboratory subjects (e.g. engineering, physical sciences)	2	1.6	1.5	1.5	1.7	2.1	1.75	2.5	2.4

Note: Univ. = University; UC = University College ; UAS = University of Applied Sciences; Poly. = Polytechnic. Table adapted from OECD (2022)

Table 1 illustrates how different funding systems apply weighting factors to account for these cost variations. As discussed in the OECD's analysis for Portugal (OECD, 2022), the distribution of cost factors shown for Portugal's allocation formula is broadly consistent with those used in comparable OECD jurisdictions, although the two Nordic systems included use a narrower span of values (e.g. Denmark and Finland use three categories for universities). While Portugal uses a comparatively large number of cost categories, particularly for universities, it is not uncommon for systems to use different levels of detail (OECD, 2022; OECD, 2021). Crucially, evidence and economic logic indicate that when funding formulas fail to adequately reflect these higher costs – a challenge highlighted in analyses by the OECD and EUA (Pruvot & Estermann, 2022) – this creates significant systemic disincentives. Such disincentives can lead to universities limiting increases in enrolment into high-demand STEM fields, despite the societal need for such skills (See, for example, Murnane et al. (2024), who note supply constraints in healthcare versus STEM). They also lead to challenges in maintaining the state-of-the-art equipment that is crucial to quality education, or may lead to institutions implicitly relying on cross-subsidisation from less expensive fields, thereby potentially straining resources across the institution. Understanding and addressing these cost differentials in funding mechanisms is therefore a prerequisite for any strategy that aims to significantly expand capacity in STEM.

These insights are crucial for policymakers seeking to design funding systems that are efficient, equitable and responsive to national priorities such as building capacity in STEM (De Witte et al., 2019). Key considerations involve balancing stable core funding (essential for long-term planning and infrastructure) with performance incentives (which can drive specific outcomes, but requires careful design to avoid unintended consequences); ensuring that funding mechanisms transparently account for varying disciplinary costs (as discussed above); and understanding the potential behavioural responses of universities to various funding signals (e.g. avoiding expansion if underfunded). This findings of the present report with regard to effective interventions must be viewed alongside these funding realities.

Simultaneously, the availability and structure of student financial aid, including needs-based grants and merit scholarships, vary considerably across the EU. This variation may create disparities in access, particularly for students from lower socioeconomic backgrounds, who may be more debt-averse or reliant on upfront support (Kumar & Sahoo, 2021; Mau & Li, 2018; Wang, 2013). Policies regarding the generosity, targeting (needs-based vs. merit-based, potentially STEM-specific) and form (grants vs. loans) of financial aid are therefore critical levers that influence not only the overall number but –fundamentally – the socioeconomic diversity of students entering STEM fields. Effective aid systems can mitigate financial barriers that might otherwise prevent talented individuals from pursuing costly STEM degrees.

Beyond direct financial considerations, the concept of “science capital”, which encompasses an individual’s science-related cultural and social resources, attitudes, and experiences, plays a significant role in shaping engagement. Research carried out in England by DeWitt & Archer (2017) highlights that the accumulation of science capital, which is often strongly linked to family background and out-of-school experiences, is unevenly distributed across different social groups. This creates systemic inequities in young people’s access to and participation in the broader STEM learning ecosystem, suggesting that those with initially higher science capital (“the rich”) tend to engage more often in enriching informal science learning experiences, thereby accumulating further capital (i.e. they “get richer”). Understanding these mechanisms is crucial when considering how to create more equitable STEM pathways across the EU.

Furthermore, national labour market conditions significantly influence the perceived attractiveness and long-term viability of choosing a STEM career (Saunders-Smiths et al., 2021). High potential earnings in STEM can act as a strong pull factor – although interventions providing wage information show complex gendered effects (Ding et al., 2021). Furthermore, perceptions about demanding work cultures, rapid skills obsolescence leading to flatter wage profiles (Deming & Noray, 2018, 2020), a lack of work-life balance, insecure contract work, or gender bias within specific industries can be significant deterrents that disproportionately affect women’s choices (Ceci & Williams, 2011; Kahn & Ginther, 2017). Therefore, policies aimed at improving workplace

conditions within STEM sectors – promoting flexibility, supporting continuous professional development, ensuring equal pay and career progression opportunities, providing adequate and affordable childcare support, and actively combating gender bias – are complementary and crucial to making STEM careers appealing and sustainable, and especially for retaining female talent beyond graduation. Issues relating to a sense of belonging, such as the feeling that one has a valued place within the STEM profession, also play a critical role, particularly for underrepresented groups (Cannaerts et al., 2025; Veldman et al., 2021).

At EU level, initiatives such as Horizon Europe (which funds research and innovation), the European Education Area (which promotes collaboration and mobility) and targeted projects such as “Girls Go Circular” (which uses engaging themes such as the circular economy, combined with mentorship and digital skills training, to specifically boost girls’ interest and confidence in STEM) aim to foster a more conducive environment (also see Langie & Pinxten, 2018 on transition projects). Furthermore, as a practical acknowledgement of the higher operational costs in many STEM disciplines, specific EU funding mechanisms such as the Marie Skłodowska-Curie Actions (MSCA) grants, part of Horizon Europe, often include provisions such as top-up lump sums to help cover laboratory expenses. This represents an example of good practice in supporting STEM research and training. Recent national reforms, often linked to NextGenerationEU recovery plans, frequently include adjustments to higher education funding models, potentially incentivising shifts towards the use of performance indicators, graduate employability, or alignment with national strategic priorities such as the digital and green transitions, which indirectly impact the focus and provision of STEM (Pruvot & Estermann, 2022). However, the effectiveness of these broad strategies hinges on their tailored implementation within diverse national contexts, as well as on rigorous evaluation – for example, Lynch et al. (2019) highlight the features of effective PD/curriculum programmes, while Broos et al. (2017) explore learning analytics dashboards.

In presenting the findings of this report, it is essential to acknowledge this complex systemic backdrop, as it is within this landscape that we aim to provide vital insights for policymakers as they seek to build a more prosperous, equitable, and innovative future for all EU citizens. To achieve this aim, the report proceeds in three key stages. First, it outlines and analyses the range of factors known to influence STEM career choices, drawing on the wide range of evidence now available while distinguishing carefully between correlational and causal variables, particularly within the EU context. Second, the report reviews evidence on diverse intervention designs aimed at improving participation and success in STEM. Lastly, the report synthesises these research findings into actionable policy recommendations that policymakers can implement towards an effective and equitable future for STEM in the EU. Before presenting these three key stages, the next chapter of the report first outlines the methodology used for the literature review.

2. Methodology

This chapter details the methodology employed to gather, evaluate and synthesise the evidence presented in this report.

2.1. Literature search strategy

To identify relevant academic literature, a comprehensive search was conducted across a range of databases. These included ERIC, PsycINFO, Web of Science, Scopus, Google Scholar and EU-specific platforms such as CORDIS, which indexes EU-funded research projects. In addition to traditional databases, AI-powered tools such as Elicit.org, Semantic Scholar, OpenAlex, Connected Papers and Research Rabbit were used to complement the search process. These tools helped to refine queries, identify citation relationships and highlight studies on the basis of their methodological rigour and relevance.

The search employed combinations of keywords related to core concepts including the field (STEM and its constituent subject areas); educational stage and choice process (terms such as education, choice, pathway, participation, enrolment, entry); specific interventions (programme, policy, experiment); key demographic groups (e.g. gender, socioeconomic status); relevant outcomes (enrolment, intention); and geographical focus (European Union and specific Member States). This multi-pronged search strategy aimed to capture a wide array of potentially relevant studies for subsequent screening.¹

2.2. Inclusion and exclusion criteria

Following the initial search, a careful selection process was undertaken to ensure that the studies included in the review provided the most relevant and reliable evidence to inform policy on participation in STEM. The primary focus was on research that examines the factors influencing students' decisions to enter STEM education at levels of secondary or higher education, including their aspirations, intentions and actual enrolment decisions. Consequently, studies concentrating solely on the retention of students after they had already embarked on STEM studies were excluded.

Furthermore, priority was given to studies offering insights directly applicable to the EU. Of these, research conducted within EU Member States was considered

¹ An example search string used during the literature review was: (STEM OR "science education" OR "technology education" OR "engineering education" OR "mathematics education") AND (choice OR selection OR pathway OR participation) AND (intervention OR program OR policy OR experiment) AND (gender OR women OR race OR ethnicity OR "socioeconomic status").

most relevant. However, significant studies from outside the EU, such as the United States, were also included if they offered valuable lessons about influencing factors or intervention designs that are potentially transferable to the European context. In such cases, the report explicitly discusses the potential limitations of applying these findings across different educational and cultural systems.

An important filter was the quality and nature of the evidence. Strong preference was given to studies capable of demonstrating cause-and-effect relationships – primarily, those using rigorous experimental methods such as randomised controlled trials (RCTs), or quasi-experimental approaches such as difference-in-differences (DiD), regression discontinuity designs (RDD) and instrumental variables (IV). Such designs provide greater confidence that the outcomes observed are attributable to the factor or intervention being studied. Well-conducted observational studies, especially longitudinal research tracking individuals over time and employing strong statistical methods to account for other influences, were also considered to be valuable sources of information, though their findings with regard to causation are interpreted more cautiously. Conversely, studies that did not present empirical data, such as opinion pieces or purely theoretical articles, were excluded, as were studies with significant methodological weaknesses or which provided insufficient detail to allow their validity to be properly assessed. Lastly, only studies whose full-text versions were readily available were included in the synthesis.

2.3. Conceptual scope: skills gap vs. labour gap

When evaluating the factors and interventions that influence participation in STEM, it is useful to distinguish between two related but distinct challenges: the labour gap and the skills gap. The labour gap refers to a quantitative shortage of individuals with specific STEM qualifications needed to fill available jobs in the economy (e.g. a shortage of software engineers or chemical technicians). Interventions whose primary aim is to increase enrolment in specialised STEM degree programmes – which are the main focus of this report, as per the inclusion criteria centring on choice and entry – directly address the labour gap by increasing the supply of qualified professionals. The skills gap, meanwhile, refers to a broader deficiency in STEM-related competencies (e.g. digital literacy, data analysis, critical thinking, problem-solving) across the wider workforce, including graduates from non-STEM fields. Addressing the skills gap might involve integrating basic STEM concepts or skills into non-STEM curricula, promoting lifelong learning in STEM areas, or fostering interdisciplinary programmes – approaches that largely fall outside the primary scope of this review unless directly linked to initial pathway choices.

While the primary focus of this review is on interventions aimed at increasing enrolment in STEM programmes (thus addressing the labour gap), it is important

to recognise that some of the interventions reviewed (e.g. improving school STEM education, as discussed in Section 3.1) could potentially enhance general STEM literacy (addressing the skills gap) alongside encouraging more students into specialised STEM pathways (addressing the labour gap). Understanding this distinction helps to clarify the report's focus and allows a more nuanced assessment of the potential impacts of the interventions evaluated.

2.4. Data extraction and analysis

For each study selected through this screening process, relevant information was systematically extracted to allow comparison and synthesis. The key details recorded included the study's authors; its year of publication; the research design; the characteristics of the people studied (such as their age, educational level and location) and the group size; specific details of any intervention being evaluated; the main outcomes measured; the primary findings (including the size of any observed effects, where reported); any available information on the costs associated with interventions; and any limitations acknowledged by the original authors.

The information extracted formed the basis for a thematic synthesis. Findings were organised around those key factors that were identified as influencing STEM choices (such as family background, psychological attributes or school experiences) and around the different types of interventions tested (e.g. role model programmes, changes to teaching methods or financial support schemes).

Throughout the course of the analysis, the quality of the evidence was critically appraised. The analysis gives greater weight to conclusions that are based on findings from studies with strong causal designs – especially RCTs and quasi-experimental research – clearly distinguishing these from correlational or descriptive findings. The limitations inherent in non-causal studies are explicitly acknowledged when discussing their results.

Lastly, the present report has aimed to provide a comparative overview of interventions. Where data permitted, this involved assessing both the effectiveness of the various approaches (i.e. the magnitude of their impact on STEM participation) and, ideally, their cost-effectiveness. These were used to categorise interventions on the basis of their evidence profile and potential feasibility. Recognising that detailed information on cost is often scarce in research literature, the analysis also includes qualitative comparisons based on whatever cost data could be gathered. This structured appraisal process seeks to equip policymakers with a clear understanding of what works, for whom, and under what conditions – based on the most reliable evidence currently available.

2.5. Limitations of the literature review

In addition to its strengths, the limitations of the literature review should also be acknowledged. First, despite making use of multiple databases, AI tools and a comprehensive keyword strategy (described in Section 2.1), the search may not have captured every relevant study. Literature published exclusively in languages other than English (or the other primary languages included in the search) may have been missed. The same may apply to unpublished reports, dissertations, conference proceedings or studies that use highly specific or alternative terminology. Moreover, because this report concerns a narrative review, it is not intended to be exhaustive.

Second, the academic literature itself is subject to potential publication bias. Studies that find statistically significant positive effects are often more likely to be submitted and accepted for publication than those which report null or negative findings. This so-called “file drawer problem” – namely, that null or negative findings may be “filed away” without being published – could potentially lead to the effectiveness of certain interventions being overestimated, despite the existence of a body of contradictory evidence, which remains unpublished. While a meta-analysis could mitigate this issue, this is beyond the scope of the present report.

Third, while the inclusion criteria used for this review are necessary for focus, they inherently limit its scope. The review’s focus on factors that influence choice and entry into STEM in higher education means that studies which solely investigate retention or persistence after enrolment were excluded – even though such studies might offer complementary insights. Furthermore, the strong prioritisation of causal evidence (RCTs, robust QEs) means that potentially valuable contextual details or hypotheses generated by purely qualitative or descriptive studies might be less prominent in the synthesis, even though context is crucial to policy implementation.

Fourth, it remains challenging to assess the direct transferability of findings from one context to another – particularly those from non-EU countries (such as many US-based studies) to the diverse national systems within the EU. While the present report makes efforts to explicitly discuss relevance and potential limitations, differences in cultural and economic contexts, education systems and the labour market mean that an intervention that is effective in one setting is not guaranteed to work identically elsewhere without careful adaptation and local evaluation.

Lastly, as explicitly noted throughout the report (e.g. in Section 2.3 and Section 5.3), one significant limitation is the scarcity of robust, comparable data associated with the cost-effectiveness of the interventions evaluated. This limits our ability to make definitive statements about the relative efficiency of different

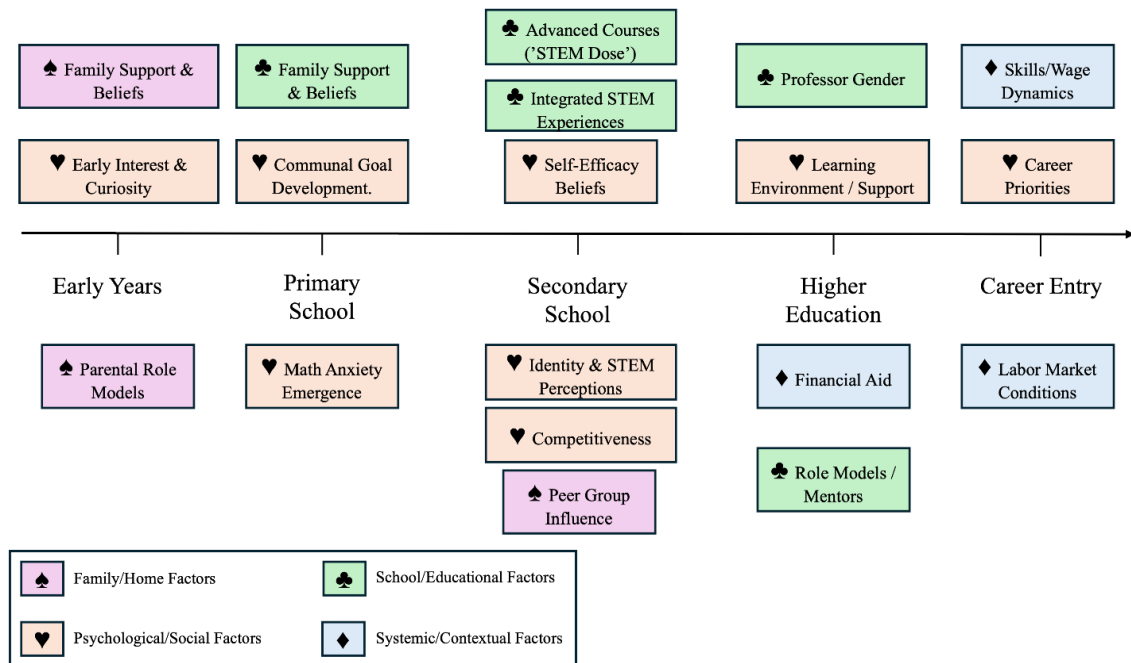
approaches and, for many interventions, necessitates a reliance on qualitative cost estimations.

3. Factors influencing STEM career choices

To develop effective and targeted policy interventions, it is essential to first understand the multifaceted factors that shape the decisions of students to pursue STEM education and careers. An individual's journey towards a STEM career is influenced by the complex interplay of factors operating at different stages in their life. A systematic literature review by Zhou and Shirazi (2025), which synthesised 159 studies mostly from the USA, Turkey and South Korea, emphasises the multi-dimensional nature of these influences. Typically, they are categorised into individual, socio-contextual (family, school, peers) and broader societal factors, whose relative importance varies across educational stages and between cultural contexts.

This chapter of the report categorises these influences into three broad stages: early influences, psychological and social factors, and post-secondary experiences, reflecting the structure visualised in the timeline in Figure 1. This figure provides an overview of the key factors identified in the literature review and discussed throughout this chapter. The figure plots these along an approximate developmental timeline from early years through to career entry, indicating the points at which they often become especially salient. The factors included represent the major themes and significant variables that are consistently highlighted in the research synthesised for this report. To make the categorisation clear and accessible, each factor in Figure 1 is distinguished by both a colour and a corresponding symbol, as explained in the legend. This coding system helps to illustrate the diverse origins of the factors that influence STEM pathways. The vertical placement of factors within each stage (either above or below the timeline) is primarily for visual clarity and grouping, and does not imply a conceptual distinction such as positive versus negative influences. These early years significantly shape an individual's long-term interests and capabilities (Wai et al., 2010). Experiences during childhood and adolescence – particularly those concerning family interactions, educational opportunities and, potentially, inherent predispositions – lay the groundwork for future STEM engagement (Besecke & Reilly, 2006; Wai et al., 2010).

Figure 1. Timeline showing the approximate stages at which key factors that influence STEM choices become salient



Note: the figure includes a colour-coded legend indicating the categories in which each factor belongs (bottom-left corner).

This chronological and thematic structure allows us to explore how factors operating at different stages of life interact and shape STEM pathways.

3.1. Early influences

Family support and role models

This section examines the impact of the home environment on an individual's STEM aspirations and choices. This includes parental encouragement, beliefs, socioeconomic status and the presence of family members as role models.

Family environment plays a crucial role – although much of the evidence that identifies specific factors is correlational rather than causal. For instance, qualitative research (structured email interviews with 30 diverse scientists) highlights **family support and a lack of discouragement as key career influences**, thereby offering rich insights but not establishing causality (Besecke & Reilly, 2006). Other non-causal studies **reinforce the importance of specific family members and related figures**. A cross-sectional survey from Slovenia found associations that suggest mothers – as well as educators perceived by students as “good teachers” (a subjective assessment) – have significantly **influenced the STEM choices of girls more than those of boys** (Cerinsek et al., 2013). Similarly, another quantitative cross-sectional survey from Mexico **identified mothers or female guardians as the family figures that were most influential over career choices** (Avendaño Rodríguez et al., 2020).

Furthermore, correlational evidence from a large-scale cross-sectional survey in the US shows an **association between having parents in STEM occupations and students choosing STEM majors** (Moakler & Kim, 2014). Meanwhile, a longitudinal causal study from Switzerland identified **parental support as being decisive among female students pursuing STEM** (Bieri Buschor et al., 2014). Reinforcing the predictive importance of parental support, non-causal longitudinal survey data from the US indicates that parental STEM support – operationalised through actions such as encouraging STEM courses, discussing STEM careers or facilitating STEM activities – is particularly crucial in maintaining the STEM career ambitions of students whose parents lack college degrees (Starr et al., 2022). While these studies consistently highlight the association between family support/family factors and STEM choices, direct causal evidence primarily comes from intervention studies.

The influence of an individual's home environment extends beyond direct encouragement to include what DeWitt & Archer (2017) term “science capital”. Their study of nearly 6,000 secondary school students in England found that higher levels of science capital – which includes components such as parents' scientific knowledge, family discussions about science and engagement in science-related activities outside of school – were strongly associated with a greater likelihood of student participation in various informal science learning experiences. This finding underlines how family-provided resources and cultivated dispositions contribute significantly to early STEM engagement, with disparities in science capital often reflecting broader socioeconomic inequalities. Students from backgrounds with higher cultural capital, as well as students with family members who work in science, reported greater participation in informal, everyday and school-led science enrichment activities.

Crucially, interventions targeting parental beliefs can yield concrete results. A US-based field experiment involving 188 adolescents (predominantly European American, college-bound high school students graduating around 2010) and their parents provided causal evidence that **students whose parents received information highlighting the usefulness and value of STEM courses subsequently took significantly more semesters of science and mathematics, compared with a control group** (Harackiewicz et al., 2012). This suggests that shaping parental beliefs regarding the value of STEM can directly influence student behaviour. The intervention itself, which focused on the provision of information (brochures and a website), appears potentially low-cost and scalable. However, caution should be applied with regard to the direct transferability of the intervention to diverse European contexts, given cultural variations in parent–child dynamics and educational systems. Building on this, family influence could be fostered by not only providing information but also by encouraging informal family activities that trigger an interest in science from an early age, as suggested by Avendaño Rodríguez et al. (2020).

School experiences

This section focuses on the role of formal and informal educational experiences prior to university in shaping STEM readiness and interest. These include advanced coursework, extracurricular activities, specific teaching approaches and targeted programmes.

Experiences during primary and secondary education are critical. A 25-year longitudinal study in the US found that **a higher density of advanced pre-collegiate STEM educational opportunities (a higher "STEM dose") is strongly associated with notable STEM accomplishments later in life** for mathematically talented adolescents (Wai et al., 2010). While this large-scale study shows a powerful correlation, it is not strictly causal. Similarly, high school maths achievement and exposure to maths and science courses are significant predictors of a student's intent to major in STEM, as well as their subsequent entry into a STEM field (Wang, 2013; Pinxten et al., 2019b).

Interventions during primary and secondary education can be impactful. A meta-analysis of 95 experimental and quasi-experimental studies found that **STEM professional development for teachers and curriculum programmes across primary and secondary education yielded an average positive impact on student outcomes**. This was particularly true when such interventions helped teachers to use curriculum materials effectively, when they focused on teachers' content knowledge, and when they included collaborative elements such as summer workshops and teacher meetings (Lynch et al., 2019). Specific curriculum approaches also show promise. Studies in Belgium found that an **integrated STEM curriculum in secondary school related positively to students' interest in STEM, its perceived relevance, the application of concepts, and engagement** (Knipprath et al., 2018; De Meester et al., 2015; Thibaut et al., 2018). An RCT in Italy involving 710 secondary school students provided causal evidence that creative STEM activities delivered via FabLabs (small-scale workshops offering tools for digital fabrication using a hands-on approach) significantly enhanced students' determination (defined as perseverance and passion for long-term goals) and externally assessed creativity (Ballerini et al., 2024). Furthermore, quasi-experimental evidence from the US suggests that **attending STEM-focused high schools** that are inclusive – meaning they typically have open enrolment policies designed to serve diverse student populations, rather than being highly selective – **increases advanced maths/science course completion rates, STEM activity involvement and interest in science careers** (Means et al., 2016). **STEM-focused high schools are characterised by requiring advanced STEM coursework, providing student-centred instruction, and offering ample support**. Other US quasi-experimental work has found that an integrated life sciences and engineering curriculum improved middle school students' science learning and retention (Anwar et al., 2022).

Bringing together these various lines of evidence, longitudinal studies that track long-term outcomes (Wai et al., 2010; Wang, 2013), as well as intervention-based research evaluating specific programmes and curricula (Lynch et al., 2019; Knipprath et al., 2018; Ballerini et al., 2024; Means et al., 2016; Anwar et al., 2022), point towards the significant benefits of providing engaging, challenging, potentially integrated, and advanced pre-collegiate STEM experiences. While much of the evidence from large-scale longitudinal and quasi-experimental studies comes from the US, the positive findings from specific interventions in Belgium and Italy (Knipprath et al., 2018; De Meester et al., 2015; Ballerini et al., 2024) suggest that similar approaches may potentially be effective in European contexts. However, access to such enriching experiences, particularly informal science learning opportunities, is not always equitable. Research from England (DeWitt & Archer, 2017), using survey and qualitative data from 5,961 secondary school students, suggests a “rich get richer” phenomenon whereby students who already have higher “science capital” are more likely to participate in informal STEM learning. This highlights that systemic inequities in access to these foundational pre-college experiences can shape students’ STEM trajectories even before formal higher education choices are made – a crucial consideration for EU policies aimed at broader inclusion.

Personality variables

This section considers inherent or developed individual characteristics and traits, such as achievement motivation, self-esteem and (potentially) goal orientations, which might predispose individuals towards or away from STEM fields. Indeed, certain personality traits have been associated with STEM choices. Besecke and Reilly (2006), through qualitative interviews, found that women in STEM often reported a high need for achievement and high self-esteem, suggesting an internal drive to excel. However, this evidence is correlational, and the direction of causality is unclear: do these traits lead to STEM choices, or does success in challenging fields of STEM foster these traits? The authors linked these positive dispositions to what they termed “transforming experiences”, which they suggest can nurture such traits that are conducive to STEM persistence. Based on their interviews, such experiences often involved personal contact with a role model, individualised encouragement or mentoring and deep engagement with the process of scientific inquiry process (Besecke & Reilly, 2006). While the study was descriptive and not an evaluation of a specific intervention designed to build these traits in order to influence initial choice, it suggests that programmes that incorporate such elements could be beneficial. Nevertheless, there remains a scarcity of direct causal evidence from the reviewed literature on interventions specifically targeting these core personality variables in order to influence initial STEM choice.

Beyond achievement motivation, expectancy-value theory² also highlights the role of broader goals. Research indicates that gender differences in STEM interest often relate more to differing endorsements of communal goals (e.g. helping others, working with people) compared with agentic goals (e.g. power, achievement) (Diekmann et al., 2011; Wang & Degol, 2013). Experimental work by Diekmann et al. (2011) provides causal evidence that activating communal goals decreases interest in STEM, unless STEM careers themselves are framed in ways that align with such communal values. This suggests that fostering interest may involve not just building achievement-related traits, but also demonstrating how STEM pathways can fulfil diverse personal and societal goals.

Indeed, related **causal evidence confirms the impact of role models and support systems**. For instance, a large-scale field experiment (RCT) involving 19,451 French high school students found that one-hour classroom interventions by female scientists significantly increased the enrolment of high-achieving girls in selective STEM university programmes (Breda et al., 2021). Similarly, a field experiment using difference-in-differences analysis across 1,397 US university economics students showed that brief classroom visits by successful female economics majors causally increased female students' likelihood of choosing an economics major (Porter & Serra, 2020). While economics is a social science, its quantitative nature and gender imbalance make these findings on the power of role models highly transferable to STEM disciplines. Exploiting the random assignment of professors to over 9,400 cadets at the US Air Force Academy, researchers **found causally that exposure to female professors in foundational courses boosted female persistence and graduation in STEM** (Carrell et al., 2009). Follow-up analysis on ~4,700 graduates showed positive impacts on STEM Master's degrees and occupations (Mansour et al., 2020). Furthermore, a quasi-experimental study using admission lotteries for an MIT freshman learning community (ESG) found causal evidence that **participation benefitted women (Grade Point Average/credits) and minority students (major choice), suggesting that structured support and mentoring matter** (Russell, 2017). An RCT involving 133 female doctoral STEM students in the US also showed that an online **resilience training programme** (CareerWISE) which included stories of successful women, causally increased problem-solving and coping efficacy (Dawson et al., 2015).

Addressing the alignment of STEM with communal goals is also supported by direct experimental evidence. In a series of experiments involving several hundred US undergraduate students, Diekmann et al. (2011) manipulated students' exposure to different career descriptions. They causally demonstrated that **framing STEM careers to highlight communal aspects (collaboration,**

² Expectancy-value theory posits that an individual's choice, persistence and performance are determined by two main factors: (1) their expectations for success, which are their beliefs about how well they will do in a task, and (2) task value, which is their belief about how important, useful or enjoyable the task is. Students are most motivated to engage in tasks at which they expect to succeed and which they value highly (Petersen & Hyde, 2014).

societal benefit) significantly increased positivity and interest, especially among women. This suggests that communication strategies and curriculum framing that emphasises STEM's societal contributions – a factor also highlighted as important in a non-causal survey of 1,281 Slovenian STEM undergraduates (Cerinsek et al., 2013) – could be effective, low-cost tools.

3.2. Psychological and social factors

Beyond early experiences, psychological factors such as confidence, motivation and perceptions of STEM, as well as social factors such as competitiveness and belonging, significantly influence choices – particularly during adolescence and the transition to higher education.

Self-perceptions and confidence

This subsection relates to students' beliefs about their own abilities and competence, particularly in foundational areas of STEM such as mathematics, and their general academic confidence or self-concept. Confidence in one's abilities, especially in mathematics, is a recurring theme. Wang (2013), analysing a longitudinal dataset from the US based on 5,650 participants, found a correlation between students' maths self-efficacy beliefs and their subsequent intent to major in STEM, which in turn predicted actual STEM major choice. This evidence is correlational, not causal. Similarly, a large cross-sectional survey of US college freshmen found that students who had strong confidence in their mathematics and general academic abilities were more likely to choose STEM majors (Moakler & Kim, 2014). Researchers in Belgium (Van Soom & Donche, 2014) identified various profiles based on autonomous motivation and academic self-concept. They found that for male students, high academic self-concept combined with high motivation predicted better early academic achievement. National data from the US also identified science self-efficacy as one of the most important predictors of STEM career aspirations among 9th graders (Mau & Li, 2018).

Fostering self-efficacy – particularly crucial, given its strong link with STEM aspirations and choices – could involve several strategies. Designing learning experiences that ensure mastery builds foundational confidence. Providing targeted and constructive feedback helps students to understand their progress and identify areas for growth. This process is enhanced when guided by the principles of “feedback literacy” – that is, developing students' ability to actively seek, interpret and use feedback to improve their learning (Coppens et al., 2023, 2025). Furthermore, learning by observation – namely, by seeing successful peers or relatable role models succeed – can provide a powerful boost to an individual's self-belief, an effect causally demonstrated in interventions involving brief exposure to such role models (Breda et al., 2021; Porter & Serra, 2020).

Intertwined with self-efficacy is another crucial element: actively addressing the threat of stereotypes. This is especially important for underrepresented groups navigating STEM fields in which negative stereotypes about their group's abilities may persist (Kahn & Ginther, 2017). The threat of stereotypes refers to a specific predicament in which individuals feel at risk of confirming the negative stereotypes applied to their social group (e.g. on the basis of gender, race or ethnicity). Research focusing on minoritised students in the US further underlines the complexity of these influences. Durham et al. (2024), using a mixed-methods approach in a study involving 89 high school students, which highlighted the critical interplay between socio-environmental factors (such as exposure to role models and supportive social contexts) and individual social cognitive career constructs such as maths self-efficacy. Their findings emphasise that among underrepresented students, STEM aspirations are shaped not just by individual confidence but significantly by the presence of supportive ecosystems and opportunities that can buffer them against cumulative disadvantage and affirm STEM identities. Crucially, the individual does not need to believe that the stereotype is true; merely being aware that others may hold that stereotype and might judge them through that lens is enough to induce the threat. In contexts in which the stereotype is relevant – such as a challenging maths test, for women or individuals from certain minority backgrounds often stereotyped as less mathematically capable – awareness of this potential negative judgment can create significant psychological pressure.

A specific and powerful barrier that is closely related to self-perception is maths anxiety. This is characterised by feelings of tension, apprehension or fear that interfere with mathematics performance. This is more than a simple dislike of the subject; it can directly impair cognitive functions such as working memory, leading to poorer performance in tests and, critically, a behavioural pattern of avoidance (Kahn & Ginther, 2017). This avoidance effectively closes the gateway to many STEM fields. The impact of maths anxiety begins early and creates a cumulative deficit. EU-based studies show that it can interfere with the fundamental encoding of new mathematical knowledge in children as young as 6 (Tomasetto et al., 2021). This anxiety also manifests as “calculated avoidance”, whereby individuals who are highly anxious about maths are more likely to avoid effortful, high-reward maths problems in favour of easier ones (Choe et al., 2019). The influence this has on educational choices is profound. A three-year longitudinal study of Italian middle school students found that maths anxiety was one of the strongest predictors of choosing a non-STEM high school track – proving to be more influential than students' actual maths performance (Cuder et al., 2024). This link is further confirmed by research showing that maths anxiety negatively predicts the selection of maths-intensive careers. This is particularly true for female students, for whom the effect can be amplified by the threat of stereotype, creating a vicious cycle in which fear of confirming a negative stereotype fuels anxiety, thereby hindering performance and reinforcing avoidance (Eidlin-Levy et al., 2023).

Motivation and interest

Motivation and interest involve an individual's intrinsic drive, curiosity and enjoyment, as well as their perceptions of the value or relevance of STEM subjects and potential careers.

Intrinsic motivation and interest are powerful drivers (Iterbeke et al., 2022). Cerinsek et al. (2013) found that interest in STEM subjects was a key factor for STEM undergraduates in Slovenia in choosing their field, especially for female students, who were also more influenced by lessons that showed the relevance to society of these subjects. Similarly, interest in maths was a top predictor of STEM career aspirations among US 9th graders (Mau & Li, 2018), and motivation towards science predicted interest in STEM careers among diverse US high school students (Robnett & Leaper, 2013). Beyond individual motivation, the social context of peer relationships also plays a role. Research indicates that the norms and values of a student's friendship group can significantly influence their STEM career aspirations, as individuals are often guided by the academic or career pathways prevalent within their close social circles.

An integrated STEM course – specifically, one that combines mechanics, programming and design within a unified curriculum taught in a Belgian school for 5 hours per week – was associated with higher student interest and higher perceptions of the relevance of STEM (De Meester et al., 2015). This suggests that it is crucial to foster and sustain genuine curiosity and enjoyment in STEM topics from an early age. Here, making the relevance of STEM explicit appears key. This can be achieved by connecting learning to real-world problems (Thibaut et al., 2018), highlighting societal contributions (Cerinsek et al., 2013), and using engaging pedagogies such as enquiry-based or design-based learning (Knipprath et al., 2018; Thibaut et al., 2018).

A series of five lab-based experiments involving more than 1,200 undergraduate students in the United States provides causal evidence that framing STEM careers in communal terms (emphasising collaboration and societal benefit) increases positivity and interest, particularly among women (Diekmann et al., 2011). **The degree to which students perceive a strong “fit” between their own interests and those embodied by STEM fields is also a significant factor.** A study in Belgium involving 9,162 students (Schelfhout et al., 2021) found that a better fit of interest related strongly to a student choosing STEM studies. Interestingly, this research also indicates that female students who chose STEM often exhibited an even better fit of interest with their chosen field compared with their male counterparts, suggesting that strong intrinsic interest might be a particularly powerful driver for women to overcome other potential barriers to STEM. This EU-based finding further emphasises the importance of nurturing a genuine alignment of interest.

Competitiveness

Individual differences in competitiveness can influence educational trajectories. “Competitiveness” refers to an individual’s preference for choosing and performing in environments that are competitive, relative to non-competitive ones. This tendency can influence an individual’s choices towards fields they perceive as requiring or rewarding such behaviour. From a policy standpoint, understanding the role of such preferences is significant because differences in individuals’ willingness to engage in competitive settings may contribute to the way in which students sort into various academic fields. This potentially impacts participation rates in demanding areas such as STEM, and may offer insights into persistent disparities in educational pathways, such as gender imbalances. Research from the Netherlands provides robust causal evidence connecting this trait to educational choices. In a study involving 362 secondary school students, Buser et al. (2014) experimentally measured competitiveness using a standard task. Participants first performed a task under a non-competitive, piece-rate payment scheme, then under a competitive tournament scheme. Following this, participants were asked to choose which payment scheme they preferred for a subsequent performance. This assessed each individual’s preference for competition, independently of their ability. The study found that this measure of competitiveness, which was assessed months prior to educational choices being made, strongly predicted those students who later went on to choose more prestigious, maths- and science-intensive academic tracks, even after controlling for academic ability and minimising reverse-causality concerns. The study also found that gender differences in competitiveness accounted for about 20 % of the gender gap in track choice.

However, the role of competition might differ for those already within or considering STEM. A large field experiment (RCT) involving 97,678 adults who were invited to a product development platform provides some nuance (Boudreau & Kaushik, 2023). The intervention randomly assigned participants, who were likely to receive invitations that framed their participation on the platform as either competitive (e.g., emphasizing contests and rankings) or collaborative (e.g., emphasizing teamwork and shared goals). Among those from non-STEM fields, women were less willing than men to join under the competitive framing; however, within the subgroup identified as being from STEM backgrounds, there was no significant gender difference in response to the competitive versus collaborative framing. This causal evidence on participants’ responses to the framing suggests that significant self-selection into STEM fields may occur on the basis of individuals’ competitiveness, potentially mitigating gender differences in response to competition among this group. The findings of Buser et al. (2014), whose work was conducted in the Netherlands, are clearly of direct relevance to the EU context. Meanwhile, although the findings of Boudreau and Kaushik (2023) come from a different context, they raise important questions about self-selection that are applicable across regions. Competitiveness remains a complex trait that interacts with cultural norms, risk aversion and confidence (Kahn &

Ginther, 2017), suggesting its role may vary across different European educational cultures.

Identity and perceptions of STEM

This subsection explores how students' sense of self, including their values, goals and feelings of belonging, aligns with their perceptions about STEM fields, the work involved and the people within them. This encompasses stereotypes and the perceived compatibility of STEM with an individual's identity. Indeed, the way in which students perceive STEM fields and whether they can see themselves "fitting in" is critical. Qualitative research in Denmark has found that some students who liked STEM subjects chose not to pursue them because they perceived STEM fields as too rigid and narrow to enable them to construct desirable personal identities (Holmegaard et al., 2014). This highlights the importance of "identity alignment". Furthermore, these perceptions of rigidity can extend to anticipated work–life balance – a crucial aspect of communal and life goals. For instance, research indicates that perceptions or experiences of STEM professions being incompatible with motherhood (and parenthood more generally) can be a significant deterrent, particularly for women (Cech & Blair-Loy, 2019). Policy considerations that focus solely on immediate professional or communal goals (such as collaboration or societal impact) may overlook these crucial long-term life compatibility factors, thereby adopting too narrow a focus, which could limit the effectiveness of interventions aimed at broadening participation. The concept of "belonging" is also vital, particularly for underrepresented groups. A systematic review emphasised the complex relationship of this sense of belonging with recruitment and retention in engineering, linked to concepts such as social capital, engineering identity and self-efficacy (Cannaerts et al., 2025). Research in Belgium has shown that girls in high school actively compare their anticipated sense of belonging across different STEM fields, and that these comparisons influence their interest in specific majors (Veldman et al., 2021). This anticipated belonging can be understood as a precursor to feeling accepted and welcomed within a future professional community. This is crucial for sustained engagement, and influences initial choices.

Indeed, the development and sustaining during adolescence of a positive STEM identity – that is, the extent to which a student sees themselves as a capable "STEM person" who belongs within the field – is critical. Stringer et al. (2019) **observed that middle school students participating in extracurricular STEM programmes showed an increase in their STEM career identity over the school year, contrasting with a decline among non-participants.** For girls, having a higher initial STEM career identity was also a strong predictor of choosing to participate in these programmes, underlining the reciprocal relationship between identity and engagement.

Stereotypes also play a role. US studies have found that STEM careers are often stereotypically viewed as being incompatible with communal goals (e.g. helping people, working collaboratively). This can deter women in particular, who tend to value such goals more highly (Diekmann et al., 2011). Conversely, the early development of a positive STEM identity, sometimes since childhood, was a factor among Swiss female students who persisted in STEM (Bieri Buschor et al., 2014). The findings of Holmegaard et al. (2014) suggest that the image of STEM may need to be broadened. This could involve showcasing the collaborative, creative, communicative and socially impactful aspects of STEM work. Causal evidence supports this: experimentally framing scientific careers as being collaborative, or highlighting their communal potential, has been shown to increase positivity and interest, especially among women and those endorsing communal goals (Diekmann et al., 2011). Addressing stereotypical perceptions (Kahn & Ginther, 2017) and fostering a sense of belonging, perhaps through learning communities (Russell, 2017) or visible role models (Carrell et al., 2009; Breda et al., 2021), represent potential strategies.

3.3. Post-secondary experiences

This section of the report examines factors encountered during higher education, such as interactions with faculty, academic support systems, financial considerations, the nature of the learning environment and curriculum structure, which influence persistence, success and career decisions within STEM fields. The experiences that students have once they have entered higher education significantly impact their trajectory within STEM fields.

Academic interaction, support and the learning environment

From the outset, a student's trajectory is shaped by their initial satisfaction with their programme placement. The alignment between a student's preference and their actual enrolment can significantly influence their entire academic experience. For instance, a study of more than 1,000 medical students found that those who were placed in their first-choice programme reported being better supported, feeling less socially and academically isolated, and were more than twice as likely to intend to pursue a career in that area compared with students who received an "other preference" placement (Walters et al., 2016). While this study focused on medicine, its finding highlights a crucial mechanism: satisfaction with one's placement is linked to persistence and positive career intentions. Furthermore, EU-based research on students interested in STEM reinforces the importance of this initial match, showing that students actively seek to combine their multiple interests into a single cohesive programme, and rate the opportunity to do this as being highly important in their educational choice (Vulperhorst et al., 2018). Taken together, this suggests that students who are unable to secure a

place in their preferred programme may enter their studies with a motivational deficit, which can influence their subsequent engagement and success.

As students move into the post-secondary phase, many of the factors that influence STEM pathways relate directly to **persistence and success within a chosen programme**, rather than directly causing a student's initial choice to enter that programme. The studies discussed here largely examine the academic interactions, support structures and learning environments experienced by students already enrolled in STEM, focusing on outcomes such as achievement, skills development or retention.

Understanding these post-entry experiences is nevertheless critically relevant for strategies aimed at increasing initial STEM participation. The lived realities, challenges and successes of current STEM students significantly **shape the perceptions of these fields in the minds of prospective students**. Information about teaching quality, effective academic support (Russell, 2017; Dawson et al., 2015), peer interactions and the overall learning climate filters out through informal networks, institutional reputation and outreach. These perceptions, reflecting the experiences of those already “inside” STEM (Saunders-Smiths et al., 2021) can influence whether potential applicants view STEM fields as welcoming, manageable, and places in which they can belong and succeed (Cannaerts et al., 2025), thereby affecting the attractiveness of these pathways in the eyes of those yet to make their choice.

Positive academic interactions and supportive environments within higher education can thus indirectly influence STEM pathways. While Wang (2013) found a **correlational link** between early academic interactions and initial entrance into a STEM major in the US, much evidence in this area focuses on **persistence and success post-entry, which can indirectly shape the attractiveness of STEM fields to prospective students**. For instance, a lottery-based quasi-experimental study at MIT (USA) by Russell (2017) demonstrated that participation in a freshman learning community (ESG) – a programme offering smaller classes, alternative teaching, co-enrolment, study spaces and mentoring – causally improved women's Grade Point Averages (GPAs) and credits completed. Notably, it also **increased the likelihood of underrepresented minority students who were already enrolled choosing specific high-demand STEM majors** such as maths or computer science, highlighting how supportive environments can solidify STEM pathways **after initial university entry**.

Online interventions can also support persistence in STEM. An RCT in the US (Dawson et al., 2015) involving female doctoral STEM students found that the CareerWISE online resilience training programme causally increased participants' problem-solving skills, resilience and coping efficacy. **While this study directly reflects impacts on skills relevant to persistence among those already in advanced STEM, it does not provide direct evidence about**

whether these forms of support influence initial decisions to enter STEM higher education.

Support for non-traditional pathways, such as transfer students in Belgium, has also been studied. Here, a voluntary basic mathematics MOOC improved the maths skills of engineering transfer students (Van den Broeck et al., 2020; pre-post evidence) – although participation was not significantly linked to their overall GPA within engineering. **Crucially, this intervention focused on enhancing the skills of students already committed to an engineering pathway; it did not influence initial STEM choice.**

While not directly assessing initial STEM intake, further research from Belgium illuminates factors that are crucial to **understanding and supporting students once they have entered STEM programmes**. These studies highlight the predictive value of diagnostic tests and academic background in identifying students who are at risk of poor academic achievement (Hanssens et al., 2025, N=1973, predictive validity study; Pinxten et al., 2019b, N=1,521, cross-sectional; Van den Broeck et al., 2019, N=258–861 across cohorts, longitudinal cohort), as well as student perceptions of these diagnostic tests (Hanssens et al., 2023, N=20 interviews + 189 questionnaire, mixed-methods). They also look at how bridging students compare with traditional first-years with regard to background and achievement/dropout (Van den Broeck et al., 2018, N=~1,300–1,500 per year, comparative study); the importance of motivation, time management and concentration skills for academic achievement among enrolled students (Pinxten et al., 2019a, N=1605, cross-sectional); and the potential role of feedback literacy and self-reflection in learning for engineering students (Coppens et al., 2023, N=61–77, mixed-methods; Coppens et al., 2025, N=72–117, mixed-methods). Learning analytics tools such as dashboards can be positively received by first-year students (Broos et al., 2017, N=887 users, evaluation study), though usage may vary. **While these factors primarily affect post-entry success and persistence, positive experiences and effective support systems within STEM programmes can enhance their overall reputation, thereby indirectly encouraging initial STEM applications.**

Instructional modality and long-term career prospects also feature in post-secondary considerations. An RCT (Chirikov et al., 2020, involving 325 second-year college students from three universities) found online/blended STEM instruction yielded student learning outcomes (final exam scores, grades, satisfaction) that were equivalent to traditional in-person instruction, but at substantially lower costs. **While this study did not directly measure initial STEM choice, such cost efficiencies could improve programme accessibility or institutional capacity, thereby indirectly influencing the supply side in a way that is relevant to student decisions.**

Conversely, observational evidence from the US on long-term STEM career dynamics – such as rapidly changing skill needs and potentially flatter earnings profiles in some fields of tech – highlights factors that might influence perceptions

of career sustainability. Deming and Noray (2020; 2018) employed large-scale data from sources such as the American Community Survey (ACS) and the National Longitudinal Survey of Youth (NLSY) – from which over 11,000 individuals were involved in some analyses for the 2018 paper), as well as job vacancy data. **However, the impact of these trends on initial student choice remains speculative, depending heavily on prospective students' awareness and priorities.**

Lastly, student perceptions of the field itself, even after entry, can offer insights. Descriptive survey evidence from first-year engineering students (N=658) across Belgium, Ireland and the Netherlands (Saunders-Smiths et al., 2021) indicates that students often hold idealised, product-focused views of the profession. Specifically, they tend to over-emphasise the role of the engineer as an inventor focusing on product innovation (“product leadership”), while underestimating equally critical roles related to optimising processes (“operational excellence”) and understanding client needs (“customer intimacy”). Such narrow perceptions may also lead them to overestimate their preparedness for the profession’s diverse demands. **Such misalignments, if commonly perceived by prospective students through informal channels, could also subtly influence the pool of applicants choosing these fields and their initial expectations.**

Financial context

This refers to the influence of financial factors on the choice of a STEM major and STEM students' persistence. These include the availability of financial aid and scholarships, as well as perceptions of future earnings potential. Wang (2013) found that receipt of financial aid was associated with entrance into STEM majors among recent US high school graduates attending four-year institutions (correlational evidence). An RCT in China that tested the provision of major-specific information about wages **found that it did not significantly shift female students towards STEM, but did increase STEM enrolment among male students in rural areas** (Ding et al., 2021). This causal evidence points to a heterogeneous effect, suggesting that providing wage information can influence the choices of certain student groups (in this case, rural men), while being insufficient to shift the preferences of female students, who may weigh financial returns differently against other factors such as communal goals (Diekmann et al., 2011).

Gender of professors and role models

The presence of female faculty and role models in post-secondary settings has a demonstrable impact. This is supported by strong causal evidence. Exploiting the random assignment of students to professors at the US Air Force Academy, Carrell et al. (2009) found that **being taught by female professors in**

introductory maths and science courses significantly improved female students' performance in those courses, increased their likelihood of taking future STEM courses, and raised their probability of graduating with a STEM degree – especially in the case of high-achieving women. A follow-up study using the same context found that **assignment to female maths/science professors in freshman year causally increased the probability of high-ability female students later working in a STEM occupation and receiving a STEM Master's degree** post-graduation (Mansour et al., 2020).

Interventions involving role models beyond the faculty itself also show causal effects. A large-scale RCT in French high schools showed that **one-hour interventions by female scientists significantly increased the share of high-achieving Grade 12 girls who enrolled in selective, male-dominated STEM programmes** at university (Breda et al., 2021). This robust causal evidence from both the US and Europe underlines the powerful impact of exposure to relatable female figures in academic and professional STEM roles.

3.4. Career priorities

This section examines the values and aspirations students hold for their future careers, which influence their educational choices. Understanding what students value in their future careers provides insights into the factors that drive their choice of study. Career priorities encompass more than just financial compensation (see Section 3.3); they reflect long-term aspirations regarding work content, impact and personal fulfilment. Cerinsek et al. (2013) explored these priorities in their cross-sectional survey of 1,281 undergraduate STEM students (861 male, 420 female) in Slovenia. Their research indicated that **high priorities for all surveyed STEM students included “doing something interesting” and “doing something fulfilling using their talents and abilities”**. While some gender differences were observed, with female students tending to place a higher value in comparison to their male peers with regard to interpersonal aspects such as helping others or contributing to society (aligning with the findings on communal goal orientation noted in Section 3.2, such as Diekmann et al., 2011), the overarching emphasis on interesting and fulfilling work emerged as a common denominator for these STEM students within an EU context.

Although this evidence comes from students already enrolled in a STEM course within one EU country *and focuses on their current priorities rather than their evolution during adolescence), the findings are nevertheless valuable. They suggest that appealing to students' desire for intellectually stimulating work in which they can make use of their abilities and contribute to society, could be an effective communication strategy for promoting STEM fields. How students define “interesting” or “fulfilling” work may also intersect with the identity perceptions discussed earlier (Holmegaard et al., 2014). Furthermore, anticipated future life events can also shape career priorities and the perceived viability of certain fields.

For instance, a longitudinal study from the US by Cech & Blair-Loy (2019), which analysed SESTAT survey data from over 4,000 STEM professionals, found that parenthood significantly altered career trajectories, with **new mothers in STEM being substantially more likely to reduce their work hours or leave STEM fields than fathers or women without children**. While this study examines post-entry career paths, the anticipation of such “work–family” challenges, often disproportionately perceived by women, could influence earlier considerations of career sustainability and thus initial educational choices related to STEM, potentially shaping what they prioritise in a long-term career path. The policy implication of this is that communication efforts to promote STEM might benefit from emphasising not just job prospects but also the intrinsic interest, intellectual challenges, opportunities for meaningful impact and personal growth potential associated with these fields – aspects that resonate with values **often developed during identity formation in adolescence**.

3.5. Skills and wage dynamics

This section of the report considers how long-term career prospects, including expected earnings trajectories and the changing nature of skills requirements over time – particularly in rapidly evolving STEM fields – might influence educational decisions.

The dynamic nature of skills, particularly in technology-intensive STEM fields, is another distinct consideration. Research by Deming and Noray (2018, 2020) used large-scale US data sources such as National Longitudinal Survey of Youth (NLSY), American Community Survey (ACS) and mass job data to explore the impact of changing skills requirements on career earnings. Their analysis, based on the economic modelling of observational data, provides evidence that suggests the **initially high earnings premium associated with degrees in technology-heavy fields (e.g. computer science, engineering) tends to decline relatively rapidly** over the first decade of an individual’s career (Deming & Noray, 2020). Within their model, the authors link this causally to skills obsolescence: as technology changes, specific skills depreciate, potentially leading to flatter age–earnings profiles compared with fields in which skills demands are more stable (Deming & Noray, 2018; 2020). The authors also observed that graduates tend to sort out of faster-changing occupations as they gain experience (Deming & Noray, 2020).

While Deming and Noray present compelling evidence regarding the wage dynamics that result from skills changes within the US context, the link between this phenomenon and initial student choice remains less direct and is largely observational or speculative. It is unclear to what extent prospective students are aware of, or weigh, these potential long-term earnings trajectories and skills obsolescence risks in comparison to factors such as immediate job prospects or starting salaries. Critically, the direct transferability of these US-based findings to

the EU is limited. European labour markets, industrial structures, systems for continuous learning and wage-setting mechanisms differ significantly. Therefore, while skills dynamism in STEM is highly relevant to the EU, understanding its specific impact on career choices requires dedicated analysis using European data. Such analysis could inform policies relating to lifelong learning, skills forecasting and how STEM careers are presented, acknowledging the need for continuous adaptation alongside the opportunities offered. Moreover, the transition from STEM higher education into the labour market, particularly for women, can shape the perceived long-term attractiveness of these fields. An study from Italy by Meoli et al. (2024) analysed data on more than 42,000 graduates using logit regressions to investigate the gender gap in graduates' transition into STEM occupations. While it focuses on post-graduation outcomes, such research is crucial because the challenges or **disparities women face in translating their STEM degrees into STEM employment can influence the perceptions and subsequent HE choices of prospective female students**. If certain STEM university pathways are perceived as having a “leaky pipeline” specifically for women into related occupations within a given EU context, this could deter initial enrolment. This highlights the interconnectedness of educational choices and anticipated labour market realities.

3.6. Sorting into STEM

This section explores the mechanisms by which individuals with different characteristics, preferences (such as competitiveness) and aptitudes select or “sort”, into STEM fields versus other fields.

The processes by which individuals “sort” into particular fields such as STEM is complex. However, understanding these mechanisms is crucial to addressing underrepresentation. For example, -analysis of longitudinal population data from over 3.3 million Swedish students by Mellander and Lind (2021) examined recruitment into STEM studies. The study **highlighted the role of curriculum reforms, the flexibility of choices within the educational system, and student attitudes**. Its findings suggest that systemic factors, such as **curriculum content and the degree of flexibility there is in educational choices, have a significant impact on sorting into STEM pathways** – -an important consideration for EU Member States when reviewing their educational structures. Boudreau and Kaushik (2023) offer causal insights from a large-scale field experiment (N=97,678 adults) on an online product development platform. The authors experimentally varied the framing of participation (competitive vs. collaborative). A key finding was that while women from non-STEM fields were less willing than men to participate under a competitive framing, this gender difference was not statistically significant among individuals from STEM backgrounds. **This implies that women currently within STEM either self-select into these fields based on a different response profile to competition, or perhaps adapt their preferences and responses through exposure to**

STEM environments. On the one hand, the strength of this study lies in its randomised experimental design, which provides causal inference regarding the effect that the framing has on platform participation among this population. On the other hand, limitations exist with regard to the generalisability of the findings to initial educational choices.

The study involved a specific online task, and the population consisted of adults already possessed some field identification, rather than students making initial major decisions. While the study does not directly explain the initial sorting into STEM, it causally demonstrates that individuals who are already within STEM may have a different response to competition compared with the general population – possibly due to self-selection (as suggested by Buser et al., 2014, whose causal evidence links higher competitiveness to choosing maths/science tracks) or adaptation, as previously noted. Beyond preferences for competition, attitudes toward risk may also play a role. For example, more risk-averse students might be deterred by the perceived academic risk of failure posed by challenging and competitive STEM programmes, or by the career risks associated with rapidly changing fields. Such risks may lead them to sort into subjects they perceive as “safer” or more stable. This implies that simply reducing competitiveness within STEM environments might not be the most effective strategy for retaining women who are already inclined towards or within STEM, though it might affect initial entry or those considering STEM from the outside. Further research is needed, potentially using experimental designs within diverse EU educational settings, to better understand how factors such as competition and collaborative preferences influence the initial sorting of student groups into STEM, thereby informing interventions aimed at broader inclusion.

4. Evaluating the effectiveness of interventions to increase participation in STEM

Having examined the factors that influence STEM choices, this chapter of the report evaluates the effectiveness of various interventions designed to increase participation in STEM education pathways, particularly at the point of entry into higher education. The focus is on identifying those interventions for which there is evidence to support their positive impact on STEM enrolment or on the intention to pursue STEM studies. We critically assess the strength of the evidence, prioritising causal insights from experimental or quasi-experimental studies, and pay close attention to the relevance, transferability and potential scalability of interventions within the diverse contexts of the EU. Where available, considerations of the cost-effectiveness of interventions are also included. To provide a comprehensive overview, this chapter is structured as follows: first, we evaluate interventions that target individuals and families (Section 4.1) and those that relate to the curriculum and pedagogical approaches (Section 4.2). We then analyse the critical influence of systemic levers, specifically higher education funding models (Section 4.3) and pathways from VET into higher education (Section 4.4). Lastly, to illustrate how these different interventions are integrated into cohesive national policies, Section 4.5 presents several case studies from EU Member States, demonstrating real-world strategies for fostering STEM talent.

4.1. Interventions targeting individuals and families

Interventions that are aimed directly at students or their families seek to influence perceptions, provide information or offer support to encourage the choices of STEM subjects.

Parental involvement through utility-value information

Building on evidence that parental perceptions matter (as discussed in Section 3.1), some interventions target parents directly. The parental utility-value interventions by Harackiewicz et al. (2012) and Rozek et al. (2017) provide a powerful example. This information-based approach involves informing parents about the utility of STEM, and was found to causally demonstrate an increase in students' decisions to take STEM courses and subsequently pursue careers in STEM. The approach is notable for its potentially low cost and high scalability, making it a promising candidate for rigorous testing and cost-benefit analysis in future EU-based research. The potential EU relevance of this approach lies in the general mechanism of parental influence on educational choices, which is present across diverse cultural contexts. However, successful adaptation and implementation in EU Member States would require careful consideration of

specific cultural norms around parental involvement, as well as variations in the way STEM careers and their utility are perceived in different national labour markets. It would also require policymakers to identify the most effective communication channels within varied educational systems. Adapting and rigorously evaluating such utility-value interventions within specific EU contexts is essential prior to their widespread adoption. Furthermore, a key contextual difference that might modulate the direct impact of such interventions is the national higher education funding models. In the US, where parents often directly pay tuition fees or students incur significant loans, parental concern over the “utility” and future return of educational choices may be more acute and direct. In many EU countries, where higher education is largely funded publicly through taxation, and direct parental financial outlay for tuition is often lower or non-existent, the leverage or perceived urgency of parental influence based on utility-value might operate differently. This could potentially limit the direct transferability or magnitude of the effect observed in the US study.

Role model and mentoring programmes

The presence of relatable role models is important in fostering interest and persistence in STEM (see, for example, Besecke & Reilly, 2006; Cerinsek et al., 2013). Interventions often seek to provide this by connecting students with STEM professionals or peers. Strong causal evidence supports the impact of such exposure. For example, as established in Section 3.3, female students who are assigned to female STEM professors at the USAFA demonstrated increased persistence and higher STEM graduation rates (Carrell et al., 2009; Mansour et al., 2020). Similarly, brief interventions that expose students to successful female peers or professionals have been shown to causally increase female students’ interest and enrolment in specific fields, based on evidence from both the US (Porter & Serra, 2020) and an EU-based RCT in France (Breda et al., 2021). These findings collectively suggest that exposure to successful individuals who share one’s identity can be highly impactful. Role models may also help to counter negative stereotypes about who “belongs” in STEM – although rigorous evidence is less common with regard to the impact of this specific mechanism on initial enrolment from broader role model programmes, particularly within EU contexts. Consequently, while the principle of the effectiveness of role models is strong, considerations regarding the transferability of specific US-based findings (such as the studies from the United States Air Force Academy), together with the varying availability of female faculty across EU universities, warrant careful attention.

Furthermore, many community-based mentoring programmes exist, but rigorous evaluations that establish a causal link specifically to increased STEM enrolment (rather than just improved attitudes or persistence among those already enrolled) are often lacking, particularly within the EU. While instinctively appealing, the effectiveness of various role model/mentoring programme designs in boosting initial STEM uptake requires more robust, EU-based causal evaluation. Outreach

activities such as STEM career days also seek to influence student aspirations. A large quasi-experimental study in the US by Kitchen et al. (2023), based on a nationwide retrospective cohort survey of 15,847 college students and using propensity weighting, examined the impact of attending college-run STEM career days during high school. The findings indicated that such participation had a positive impact on students' STEM career aspirations by the end of high school, suggesting that these events can be effective in sparking or reinforcing interest, although their long-term impact on enrolment requires further exploration.

4.2. Curriculum and pedagogical approaches

Changes in how STEM is taught can influence students' interest and perceptions, potentially impacting their desire to pursue STEM further.

Integrated STEM education

As discussed in Section 2.1.2, the STEM@school project in Flanders, Belgium (Knipprath et al., 2018) provides EU-based evidence (potentially causal, due to its experimental comparison design) that suggests **an integrated STEM curriculum can improve student attitudes and performance on specific integrated tasks** compared with traditional separate-subject teaching. This approach directly addresses the perception that STEM disciplines are isolated from one another. The findings from Knipprath et al. (2018) indicate that supporting the development and evaluation of such innovative curricula is relevant to EU policy, as these approaches might make STEM more engaging and conceptually coherent for a broader range of students. If such integrated approaches foster deeper understanding and sustained interest through secondary school, they could plausibly lead to increased enrolment in tertiary STEM programmes. However, there is still a need for direct causal evidence from the project's final analyses or similar follow-up that specifically links this particular integrated curriculum unit to increased tertiary enrolment. Beyond formal curricula, extracurricular STEM programmes also aim to engage students. A US-based quasi-experimental study by Stringer et al. (2019) investigated the impact of STEM-focused extracurricular programmes, using propensity score matching with 733 middle school students (71 in STEM extracurricular programmes). The authors found that participation could positively buffer STEM identity and motivation, particularly for girls, suggesting that such programmes can play a role in maintaining interest during crucial developmental years, although direct impact on later HE enrolment was not the primary outcome.

Addressing perceptions and stereotypes about STEM

Qualitative, longitudinal research from Denmark by Holmegaard et al. (2014) highlights that students' perceptions of STEM as being "rigid" or "narrow",

together with the perceived difficulty of constructing desirable identities within these fields, can deter them from pursuing STEM. In Slovenia, Cerinsek et al. (2013) also found that highlighting the societal relevance of STEM is an important factor, especially for female students. These perceptions are often intertwined with broader societal stereotypes about who “belongs” or is capable of working in STEM fields, which can disproportionately affect certain groups, including women (Kahn & Ginther, 2017).

Broader psychological literature on stereotypes provides insights into how such perceptions might influence choice. For example, Chaxel (2015) suggests that stereotypes can lead to biased processing and the distortion of information about career options, potentially steering individuals away from fields perceived as incongruent with their identity or group stereotypes. While that study does not specifically establish causality, experimental work in social psychology (e.g. Diekmann et al., 2011) shows that **activating stereotypes or highlighting communal versus agentic goals within career descriptions can causally influence interest**, particularly for women in fields often perceived as lacking communal aspects. Furthermore, challenges such as maths anxiety (Eidlin-Levy et al., 2023), which can be linked to self-perception and confidence issues that are potentially exacerbated by stereotypes, have been shown to negatively predict the maths intensity of chosen careers, especially for female university students.

Systematic reviews of interventions designed to reduce implicit biases more broadly (e.g. FitzGerald et al., 2019) suggest that **exposure to counter-stereotypical exemplars can be a promising technique** – although caution is advised regarding the robustness and real-world impact of many bias reduction programmes. Applying these insights to STEM, interventions that expose students to diverse and relatable STEM role models (as discussed in Section 4.1) or which frame STEM careers in ways that challenge traditional stereotypes (e.g. by emphasising collaboration, creativity or societal impact, as implied by Cerinsek et al., 2013, and Diekmann et al., 2011) could potentially counter negative perceptions and broaden STEM’s appeal.

An example of an EU-based intervention providing causal evidence in this area is the RCT by Grosch et al. (2022) in Austria. This found that a **digital intervention designed for elementary school children increased girls’ interest and confidence in STEM, thereby reducing the gender gap early on**. This highlights the potential of targeted interventions to address self-perceptions and confidence – key factors influenced by stereotypes – at younger ages.

Further EU-based research explores specific pedagogical interventions aimed at tackling the gender gap. One study in Spain by González-Gallego et al. (2025), which involved 879 secondary and high school students, critically examines the causes of the gender gap and investigates the influence of computational thinking projects. The authors’ analysis suggests that such projects, when applied in secondary school, **can positively influence students’ attitudes and**

potentially their progression towards STEM higher education, offering another avenue for curriculum enhancement within EU contexts. Designing and rigorously evaluating interventions that explicitly aim to broaden the perceived image of STEM through curriculum adjustments, targeted communication campaigns or confidence-building activities represents a necessary area for future research in the EU. While observational findings highlight the problem, more EU-based causal evidence is needed on effective interventions.

4.3. Systemic levers: higher education funding models

Beyond interventions which focus on individuals or pedagogy, the availability and attractiveness of STEM programmes is influenced by systemic factors that relate to how higher education is structured and funded. The European Commission has emphasised the importance of funding mechanisms (European Commission, 2022), especially given the often significantly higher costs that are associated with delivering quality STEM education compared with other disciplines. While direct causal evidence from experimental studies on the impact of funding formulas is limited, economic reasoning, stakeholder reports and comparative analyses of national systems provide strong evidence that the design of these funding systems serves as a powerful lever to influence institutional behaviour and, ultimately, the number of STEM graduates.

As established in Section 1.1, the higher education funding models of EU Member States are highly diverse. Analysis by the European University Association (EUA) confirms that allocation mechanisms for core public funding vary greatly, typically combining elements such as funding formulas, performance contracts and historical allocations in configurations that differ between countries (Pruvot & Estermann, 2022). This diversity reflects differing national traditions, policy priorities and responses to evolving pressures on public finances. One trend observed over the past two decades, which was already being noted by the OECD in the mid-2000s, has been an increasing interest in incorporating performance-based elements and the principles of New Public Management (such as greater institutional autonomy coupled with accountability) into these models. Such interest is often driven by the goals of enhancing efficiency, quality and responsiveness (Strehl et al., 2007, pp. 14, 19; Pruvot & Estermann, 2022, p. 5).

A critical issue within these models is how they account for differential programme costs. STEM fields frequently incur higher per-student expenditures due to necessary investments in laboratories, specialised equipment and consumables. They also often require lower student-to-staff ratios in order to deliver effective teaching. Many European funding systems attempt to address this through weightings applied within funding formulas based on subject cost groups, with varying levels of sophistication (e.g. seven cost groups were identified in Austria,

compared with three in Denmark, according to Pruvot & Estermann, 2022, p. 14). However, the adequacy, accuracy and regular updating of these cost weightings are crucial. **Where funding formulas are primarily driven by student numbers** (a common input indicator – see Pruvot & Estermann, 2022, Figure 3; Strehl et al., 2007, p. 11), **without sufficient or accurate cost weighting for high-cost fields such as STEM, institutions face tangible financial disincentives to maintain or expand these programmes.** As stakeholders noted in an earlier OECD work, this leads to concerns about the viability of expensive studies (Strehl et al., 2007, p. 13) or necessitates internal cross-subsidisation from less resource-intensive disciplines, potentially straining resources across the institution (Pruvot & Estermann, 2022, p. 5). Conversely, funding models that explicitly recognise and adequately compensate for the higher costs inherent in STEM education create clear positive incentives for universities to invest in quality and to expand the number of student places available, thereby directly impacting the supply side of the STEM pipeline.

Furthermore, the increasing use of performance indicators and contracts also creates differential effects on STEM (Pruvot & Estermann, 2022; Strehl et al., 2007). If performance metrics heavily favour outputs that are easily quantifiable across all disciplines (e.g. overall graduation rates) without considering the specific nature of STEM education, or if they act as an incentive to maximise student throughput, they risk inadvertently disadvantaging STEM fields or causing negative consequences on quality – a concern long raised by stakeholders (Strehl et al., 2007). The design of performance agreements therefore needs to be carefully considered to ensure they align with the goal of strengthening specialised fields such as STEM.

While direct causal evidence linking specific funding adjustments to isolated changes in STEM enrolment remains limited – hindered by the complexity of such large-scale reforms – the economic logic of such adjustments is compelling, as the structure of public funding profoundly shapes the strategic decisions that universities make with regard to programme offerings and resource allocation (Strehl et al., 2007). A key policy consideration, therefore, involves encouraging Member States to regularly review their funding mechanisms to ensure that they do not inadvertently create barriers to providing high-quality STEM education. Acknowledging such cost pressures, some EU mechanisms such as specific fellowships or parts of the Marie Skłodowska-Curie Actions (MSCA) demonstrate good practice by providing additional allowances for research costs. While such top-ups are not exclusive to STEM and depend on their justification within any project proposal (including in social sciences and humanities), they represent an acknowledgment within the funding system that resource needs can vary significantly by field. Promoting the sharing of best practices with regard to cost-informed funding models and addressing the significant gap in comparable data on funding and costs across the EU are crucial prerequisites for more rigorous analysis and evidence-based policy development in this critical area (Pruvot & Estermann, 2022; Strehl et al., 2007).

4.4. Systemic levers: pathways from VET into STEM higher education

Another critical systemic factor that influences the overall pool of potential STEM entrants is the permeability between VET systems and higher education, particularly STEM programmes. Strengthening these pathways was identified by the European Commission as a key area of interest for potentially increasing STEM enrolment.

VET systems vary considerably across the EU. In some Member States, VET is primarily geared towards direct labour market entry, with routes into university being limited or complex. In others, pathways are more clearly defined, sometimes involving bridging programmes, specific entrance exams or the recognition of prior VET learning. The ease of transition can significantly impact whether talented individuals who initially chose a vocational route can later pursue advanced STEM studies if their interests and capabilities evolve. Barriers often include insufficient academic preparation in core STEM subjects within VET curricula, a lack of clear information about transition possibilities, negative perceptions of VET qualifications on the part of universities, and complex administrative hurdles.

While the amount of rigorous causal evidence evaluating specific VET-to-HE transition policies on STEM enrolment found by this review is limited, examining existing studies does provide valuable insights into challenges and potential solutions. Conceptual work (e.g. Katartzi & Hayward, 2020, based on UK VET/HE systems and applying Bourdieu/Bernstein) highlights significant “transitional frictions” faced by VET students. These ongoing struggles arise from mismatches in habitus (learned dispositions), pedagogical regimes (teaching/assessment expectations), and differing knowledge structures between VET (often practical, context-specific) and HE (often abstract, disciplinary). These mismatches can impede students’ “epistemic access” to the ways of knowing that are valued in HE.

This theoretical understanding emphasises the importance of interventions designed to ease these frictions. Dedicated bridging programmes and forms of support aimed at filling specific knowledge gaps are crucial, reflecting the needs observed among STEM transfer students transitioning from professional Bachelor’s (similar to VET graduates) into engineering programmes in Belgium (Catterall et al., 2013; Van den Broeck et al., 2019, 2020). The above studies investigated the predictive value of background knowledge for bridging students (2019) and the effectiveness of support mechanisms such as preparatory mathematics MOOCs (2020), highlighting the need to address academic readiness. Providing VET students with targeted guidance and counselling about HE options in STEM is another implied need that arises from these transition challenges. Successfully implementing measures that reduce transitional friction – by addressing knowledge gaps and supporting students’ adaptation to the

pedagogical environment of HE – represents a significant strategic opportunity to broaden the intake into STEM higher education.

Examples of education systems that foster pathways from VET into STEM higher education

Spain

The **Spanish education system** serves as an interesting case study, as it provides an example of how vocational and higher education systems can be linked through structured, practice-oriented tertiary pathways, particularly in support of STEM participation. Central to this model are the ***ciclos formativos de grado superior (CFGS)***, or “**higher-level training cycles**”, which are part of advanced vocational education and training (*formación profesional de grado superior*). These programmes are classified as tertiary education within the Spanish qualifications framework (MECES) and the European Qualifications Framework (EQF) Level 5, and are offered across a wide range of fields, including industrial technologies, health sciences and information technology. CFGS graduates are granted direct access to related university Bachelor’s programmes, including STEM degrees, often with credit recognition for completed coursework. Universities may allow CFGS graduates to bypass certain introductory modules, thus shortening the time to degree completion. This system reflects Spain’s commitment to vertical mobility and modular progression, which enables students to build on vocational foundations without the need to return to general secondary education – an important enabler of educational inclusion and flexibility. While the Spanish system is still evolving, recent reforms such as the 2022 VET Law (*Ley Orgánica de ordenación e integración de la Formación Profesional*) have further strengthened integration between VET and higher education. These reforms emphasise dual VET (combining workplace and school-based learning), digital skills and improved guidance systems to support learners throughout their educational trajectories.

Cedefop (2023a) highlights both the strengths and challenges of the Spanish model. Although a growing number of CFGS graduates are progressing to university studies (Spanish Ministry of Education, reported by Erudera News, 2022), barriers remain in the form of regional disparities (McGowan & San Millan, 2019), inconsistent credit transfer policies (Cedefop, 2023a) and limited guidance on academic progression. Nevertheless, pilot projects such as *FP Dual Universitaria* in Catalonia and Andalusia are exploring more integrated VET–HE models, with a strong emphasis on STEM and digital fields (<https://www.formaciondualuniversitaria.org/>).

The Netherlands

The **Dutch education system** offers another example of how well-structured pathways can enhance the permeability between vocational and academic

education. A key component is the **Associate Degree (AD)**, introduced in 2006 and formally embedded in the Dutch Qualifications Framework at Level 5. These two-year programmes are positioned between secondary vocational education (*middelbaar beroepsonderwijs* or **MBO**) and higher professional education (*hoger beroepsonderwijs* or **HBO**), which is typically offered at universities of applied sciences (*hogescholen*). AD programmes are designed in close collaboration with regional employers and offer a practice-oriented curriculum that addresses specific labour market needs. What makes the Dutch model particularly notable is the seamless **vertical mobility** it enables: graduates from MBO programmes can enrol in AD programmes, and those who complete an AD can subsequently transition into the third year of a related HBO Bachelor's degree. This **stackable qualification model** allows students to progress in stages, without the need to re-enter more general secondary education – a feature that significantly reduces barriers to higher education, especially in STEM-related fields where technical AD and HBO programmes are available. Empirical studies have analysed the development and uptake of these pathways. For instance, Fazekas and Litjens (2014) discuss the integration of AD programmes into the broader Dutch tertiary education landscape and highlight the role of these programmes in increasing participation among non-traditional students. Similarly, Cedefop (2023) provides insights into the success and challenges of vertical transitions from VET to higher education, noting that while institutional cooperation is key, more support is needed to ensure retention and completion once students enter HBO programmes. From a policy perspective, the Netherlands exemplifies how **flexible learning pathways**, underpinned by clear qualifications frameworks and strong institutional alignment, can foster inclusiveness and meet both educational and labour market goals. The Dutch experience suggests that **creating intermediate qualifications like the AD**, in combination with **curricular alignment and credit recognition**, can be a powerful tool to **widen participation in STEM higher education** and promote **lifelong learning**. Overall, the Dutch model offers important lessons. Encouraging Member States to develop or strengthen **intermediate, practice-oriented tertiary qualifications** and support **transition mechanisms** – as seen in the Netherlands – could contribute significantly to the EU's broader goals of inclusive, responsive and future-ready education systems.

Finland

The **Finnish education system** stands out for its **high degree of flexibility and permeability** between VET and higher education. A cornerstone of this system is its **competence-based approach** to VET, which allows students to progress at their own pace and gain qualifications by demonstrating their skills in authentic work environments. This approach not only supports individualised learning paths but also facilitates smoother transitions into higher education, particularly universities of applied sciences (*ammattikorkeakoulut*), which play a central role in Finland's tertiary education landscape. Unlike in some other EU Member States, **VET graduates in Finland can apply directly to higher education** without needing to complete additional general education qualifications. This is

enabled by the **Finnish National Qualifications Framework**, which aligns VET qualifications with the rest of the education system, ensuring that they are formally recognised and valued. Moreover, admissions policies in many universities of applied sciences are designed to accommodate applicants from vocational backgrounds, particularly in fields such as engineering, health and information technology – key areas for expanding STEM participation. Several studies have examined the effectiveness of these pathways. Some time ago, Stenström and Lasonen (2000) emphasised Finland's efforts to ensure parity of esteem between general and vocational education – a theme reinforced by Cedefop (2019) – that signals flexible and individual learning paths and the validation of prior learning in VET. Furthermore, OECD reviews (e.g. OECD, 2020b) highlight Finland as a model for the integration of VET into the broader learning ecosystem, noting that its flexible structures reduce dropout and support lifelong learning. The case of Finland demonstrates how **policy alignment, learner-centred pedagogy** and **institutional openness** can make vocational routes a viable and respected entry point into higher education. Importantly, these pathways are not only accessible but are also designed to **support the success of students** once they are in higher education, thanks to tailored guidance and support mechanisms.

Germany

Germany provides an example of a highly structured and well-integrated system **for transitioning from VET into higher education**. The German **dual system of VET**, which combines classroom-based instruction with hands-on apprenticeship training in companies, is recognised for its strong alignment with labour market needs. While traditionally viewed as a parallel rather than a stepping-stone into academic education (Nikolai & Ebner, 2011; Deissinger, 2005), recent policy and institutional developments have significantly improved **permeability between VET and higher education** in Germany, including in STEM fields. A key feature of the German system is the presence of **dual study programmes (Duale Studiengänge)**, which combine vocational training with academic studies at **universities of applied sciences (Fachhochschulen)** or cooperative education institutions (*Duale Hochschulen*) (Euler, 2013; Graf L, 2013). These programmes enable students to earn a university degree while working part-time in a company, thus **blurring the lines between vocational and academic pathways**. Another distinct model, especially prominent in some states such as Baden-Württemberg, is that of “vocational academies” (*Berufsakademien*). These offer tertiary-level education that deeply integrates practical company-based training with academic instruction, providing a unique pathway for secondary school graduates (Deissinger, 2000). Importantly, graduates from VET programmes who do not follow a dual study programme can still gain access to higher education, provided they meet certain criteria such as vocational training certificates and work experience, a pathway that Germany has notably relied upon, often **through non-traditional entry routes**. The **German Qualifications Framework (DQR)** has further contributed to the enhancement of transitions between VET and higher education by aligning all qualifications –

including those from vocational and academic education – within a coherent structure based on learning outcomes. This enables the better recognition of prior learning and **supports lifelong learning and upskilling** opportunities. Empirical research supports the positive role of these mechanisms. Euler (2013), for example, describes the structural features of dual study programmes and their growing popularity, particularly in engineering and IT fields. Powell and Solga (2010) explore the complexities of educational transitions in Germany and highlight how permeability has increased over time, even if challenges remain in ensuring equity of access and support for non-traditional students. In addition, the OECD (2021) review of Germany's education system emphasised the success of dual routes and VET–HE linkages, particularly in maintaining low youth unemployment and responding to skills shortages. Germany's model is especially relevant for EU policy due to its demonstration of how **cooperative education models**, clear **qualification frameworks** and **professional recognition of VET credentials** can facilitate upward mobility and expand access to higher education, including STEM programmes. Encouraging other Member States to draw on Germany's experience – in particular, its integration of work-based learning with academic education – could help to advance EU goals in relation to skills development, inclusion and educational innovation.

Austria

Austria presents a distinctive and effective model for creating **permeable pathways between VET and higher education** – in particular, through its strong tradition of dual and professionally oriented education. A key element of Austria's system is the *Lehre mit Matura* programme, which allows apprentices within the dual system (company-based training and vocational school) to simultaneously prepare for and take the *Berufsreifeprüfung*, an exam that provides general university entrance qualifications. This model enables students to combine **work-based learning with academic advancement**, offering a clear and flexible pathway from VET into tertiary education. In addition, Austria has a well-developed system of **colleges for higher vocational education (Berufsbildende Höhere Schulen or BHS)**, such as the **higher technical institutes (HTLs)**. These five-year programmes offer both a secondary school diploma (*Matura*) and high-level vocational qualifications. BHS graduates are fully eligible for university admission, and many proceed to universities of applied sciences (*Fachhochschulen*), particularly in technical and STEM-related fields. The strong **horizontal and vertical alignment** between BHS, dual education and higher education institutions contributes to a highly integrated system. Austria also participates actively in the **European Qualifications Framework (EQF)** and its national counterpart, the **NQF**, which enhance transparency and facilitate the recognition of VET qualifications to enable further study. Moreover, many **Fachhochschulen** offer programmes designed to attract VET graduates, with curricular structures and teaching approaches that accommodate their prior learning and practical experience. Research highlights Austria's strengths in integrating vocational and academic education. In addition, The Federal Ministry of Labour and Economy (2022), notes that Austria's emphasis on dual and

technical education plays a critical role in preparing students for both employment and further study, especially in the sectors of engineering and IT. The OECD (2019) commends Austria for offering multiple, high-quality routes into higher education, with particular emphasis on inclusion and skills matching. Austria's experience underlines the **value of dual-purpose qualifications**, the **recognition of VET credentials for university access**, and the **design of practice-oriented higher education institutions**. These features offer important lessons for EU policy development. Encouraging Member States to adopt similar strategies, such as integrated VET programmes with academic qualifications or entrance exams tailored to VET students, could enhance participation in STEM and support more inclusive education systems across Europe.

France

France offers a set of pathways that link **VET with higher education**, in particular through its **short-cycle tertiary qualifications** and professionally oriented tracks. One of the key features of the French system is the **Brevet de Technicien Supérieur (BTS)**, a two-year post-secondary VET qualification delivered mainly in upper-secondary schools (*lycées*). The BTS is widely regarded as a **gateway to further higher education**, particularly in technical and STEM-related fields, and is taken by over 200,000 students each year. Graduates of the BTS can progress to the **Licence Professionnelle**, a one-year vocational Bachelor's programme offered by universities and **Instituts Universitaires de Technologie (IUT)**. The *Licence Pro* is explicitly designed to provide continuity for students from VET backgrounds and is often developed in close cooperation with industry. These programmes typically offer credit recognition for BTS graduates and are tailored to ensure **labour market relevance and academic progression**. France also supports **vertical mobility through the validation of prior learning (Validation des Acquis de l'Expérience, VAE)**, allowing individuals with work experience or informal learning to access higher education without traditional academic credentials. The **French Qualifications Framework (Cadre National des Certifications Professionnelles or CNCP)** facilitates these transitions by aligning VET and higher education qualifications and improving transparency. Academic research has examined the role of the BTS and *Licence Pro* programmes in expanding access to higher education and fostering professional identity. Cedefop and Centre Inffo (2023) highlight the importance of curriculum articulation (ensuring a logical connection between VET and HE curricula) and providing student support in ensuring successful VET-to-HE transitions. Furthermore, comparative research suggests that these vocational pathways in France can play a significant role in promoting equity; for example, Imdorf et al. (2017) found that such programmes help to increase access to higher education for women from less privileged social backgrounds in particular, contrasting with patterns observed in other national contexts. Meanwhile, policy reports such as OECD (2020a) note that France's short-cycle tertiary sector plays a critical role in **broadening access to STEM education**, especially for students from less academic backgrounds. The French model

demonstrates that **bridging mechanisms**, such as post-secondary VET qualifications with clear progression routes and industry relevance, can be effective tools for increasing participation in STEM. The French experience illustrates the importance of investing in **professionally oriented tertiary education** and ensuring that VET learners can **transition upward without dead ends**. Promoting such mechanisms – through shared curricula, credit recognition and flexible admission systems – could help to diversify and expand the STEM talent pipeline.

However, it is important to note that while effective VET-to-HE pathways, such as those in France, can significantly broaden access and increase the diversity of entrants into STEM higher education, a country's overall share of STEM graduates is influenced by a wider array of factors. For instance, countries such as Germany and Austria, which report high shares of STEM graduates (36 % and 30 %, respectively, compared with some countries, in which the share may be as low as 15–17 %) benefit not only from specific pathway structures but also potentially from strong industrial demand, overall STEM capacity in HE, and distinct educational system traditions. Thus, while the French model offers valuable lessons regarding pathway permeability, achieving a high national output of STEM graduates requires a comprehensive strategy that addresses multiple influences across the educational and societal landscape.

Czech Republic

The Czech Republic offers another example from Central/Eastern Europe in which VET pathways are well established and popular – although transitions into higher education (HE) present a mixed picture. A significant proportion of students enrol in upper-secondary VET programmes. Typically, four-year programmes culminate in the *Maturita* examination, which grants access to tertiary education, including STEM fields (OECD, 2020c). For those students who complete shorter (two- or three-year) VET programmes, bridging courses are generally required in order to enter HE, and these routes have been associated with lower subsequent HE graduation rates (OECD, 2020c).

The OECD (2020c) notes that while VET is popular in the Czech Republic, ongoing efforts are needed to strengthen its quality and labour market relevance to improve graduate employability (ibid, p. 3). Initiatives such as “The Modernisation of VET through the Collaboration with Industry project and the National Register of Qualifications aim to better align VET curricula with the needs of employers and with national qualification standards (OECD, 2020c). However, systemic challenges, including early educational tracking and regional disparities, can impact equitable access and progression (OECD, 2020, p. 3). The example of the Czech Republic highlights the importance of robust VET systems as a potential feeder into HE, but also underlines the need for continuous improvements in quality, strong links with industry and well-supported bridging mechanisms to ensure that VET graduates can successfully transition into and complete higher education programmes in STEM.

4.5. National strategies and case studies

Having evaluated the effectiveness of various types of interventions – from targeting individuals to reforming systemic levers – it is clear that no single action can solve the challenge of increasing participation in STEM. To be effective, policies must combine multiple, evidence-informed approaches into a coherent national strategy. The case studies that follow, from Ireland, Sweden, Germany and Spain, serve to illustrate how various EU Member States are building such comprehensive ecosystems. They provide real-world examples of how the factors and interventions discussed in this report are being strategically addressed at national level, highlighting both common priorities and context-specific solutions.

Ireland

Ireland provides a compelling case study of a coordinated, national strategy that aims to create a “whole-of-system” STEM ecosystem, as outlined in its STEM Education Policy Statement 2017–2026 (IDA Ireland, 2021). The Irish approach translates evidence into a portfolio of targeted actions across the entire educational pipeline. This policy framework is brought to life through a dense network of more than 70 distinct programmes and initiatives, which can be categorised according to three key themes.

Targeting individuals and families through role models and engagement

A significant portion of Ireland’s strategy focuses on early engagement and exposure to role models, with the aim of addressing biases and building confidence. Initiatives such as I WISH, SciFest (Ireland’s largest STEM fair programme, with around 60 % female participation) and Teen-Turn directly connect students with the world of STEM through outreach, competitions and mentorship. The aim of these activities is to influence course selection and to counter stereotypes. Programmes such as the “BT Young Scientist and Technology Exhibition” provide a high-profile platform for young people, including girls, to showcase their abilities. This aligns with evidence from Breda et al. (2021) and Porter and Serra (2020) on the causal impact of exposure to role models. Furthermore, programmes such as Smart Futures by Science Foundation Ireland explicitly provide STEM career resources for students, teachers and parents, echoing the effective parental information strategy identified by Harackiewicz et al. (2012).

Enhancing pedagogical and pre-college experiences

Ireland’s approach emphasises hands-on, enquiry-based learning, which has been associated with increased student interest. Programmes such as Coder Dojo, Learnit (in partnership with LEGO Education), and VEX Robotics provide informal, creative and often collaborative environments for students to develop

technical skills, addressing the need for engaging pre-college experiences discussed in Section 3.1. The STEAM-ed programme, which uses a co-teaching model that pairs primary school teachers with industry experts, directly enhances teachers' capacity while enriching the student learning experience. The STEM Teacher Internship Programme further deepens this connection by placing pre-service teachers in industry, enriching their pedagogical content knowledge with real-world relevance.

Post-secondary and career support: a focus on retention and pathways

The Irish ecosystem extends well beyond initial attraction. In the post-secondary and professional space, initiatives focus heavily on retention and on creating alternative pathways. University-linked programmes such as CodePlus (Trinity College Dublin) and EsTeEM Mentoring (Technological University Dublin) provide crucial support and mentorship for female students who are already enrolled in engineering and computer science. This aligns with the importance of learning environments and a sense of belonging identified in Section 3.3. Critically, initiatives address the need for diverse career pathways and support for individuals re-entering the workforce. Programmes such as Women TechSTART and Women Reboot offer fully funded training and work placements for women from non-technical backgrounds or those returning from a career break to transition into technology roles. These initiatives create new talent pipelines, similar in spirit to the VET-to-HE pathways discussed in Section 4.4, by recognising that talent exists outside traditional, linear educational tracks.

Sweden

Sweden's approach is characterised by both a forward-looking national strategy and an insightful, evidence-based analysis of its own past policy experiments. This dual focus is especially valuable, as it demonstrates how a country can learn from its history to design more effective interventions in the future.

A future-oriented national STEM strategy

In 2025, the Swedish government launched a comprehensive STEM strategy spanning from preschool to postgraduate study, with the explicit goal of reinforcing Sweden's position as a leading technology nation (Ministry of Education and Research, 2025). Motivated by a dip in PISA mathematics scores and a decrease in the number of pupils choosing science and technology tracks, the strategy sets ambitious quantitative targets:

- Increase the share of top performers in mathematics in PISA to at least 15 % by 2033.
- Increase the share of upper-secondary pupils choosing science and technology programmes to 25% by 2035.

- Increase the number of full-time students in higher education STEM fields to at least 90,000 by 2035.

A key component of this strategy is the creation of a new STEM delegation, chaired by a prominent female leader from the technology industry. This delegation is tasked with improving coordination between public actors and the business sector. Crucially, it has a specific focus on increasing the interest of girls and women in STEM. This move directly embraces the importance of visible, high-level role models to influence perceptions and aspirations – a mechanism for which strong causal support is identified in this report (Breda et al., 2021; Carrell et al., 2009).

Lessons from past reforms: the Importance of incentives and flexibility

A retrospective analysis of major Swedish upper-secondary curriculum reforms by Mellander and Lind (2021) provides critical lessons on how to increase participation in STEM effectively. Their research, based on longitudinal register data spanning three decades, compares two major reforms and concludes that policies based on positive incentives and flexibility are likely to be more successful than those based on requirements.

For example, a 1995 reform that aimed to make vocational track more attractive by requiring vocational students to take more academic STEM subjects paradoxically decreased enrolment in those vocational programmes, and did not increase progression to university. This contrasts sharply with the success of a supplementary one-year upper-secondary science/technology education programme. This voluntary “add-on” programme, which targets students who have already completed a non-STEM academic track, offers them a guaranteed university place in a STEM field upon completion. This low-risk, high-reward pathway has proved exceptionally effective in boosting STEM recruitment, particularly among female students (Mellander & Lind, 2021).

This finding underlines the power of creating flexible and alternative pathways into STEM. This is highly relevant to the VET-to-HE transition mentioned previously in this report. The Swedish experience suggests that such “bridging” or “add-on” programmes that lower the perceived risk of committing to a demanding STEM track can successfully attract a wider and more diverse pool of talent.

The case of Sweden demonstrates a dual approach: a clear, ambitious future strategy combined with deep learning from past policy experiments. Key takeaways for the EU include:

- **The power of positive incentives and flexible pathways:** voluntary “add-on” or bridging programmes that offer high rewards (such as guaranteed university entry) can be more effective than compulsory measures.
- **The importance of a long-term, coordinated focus on gender diversity:** the establishment of a dedicated delegation with female

leadership signals a serious commitment to addressing underrepresentation.

- **The need for evidence-based policy:** Sweden's use of longitudinal data to evaluate large-scale reforms provides a model for understanding the real-world (and sometimes unintended) consequences of policy decisions.

This combination of forward-looking ambition and reflective, evidence-based adjustment makes Sweden a key case study for other EU Member States aiming to design effective and sustainable STEM policies.

Germany

At the heart of Germany's strategy is a comprehensive national framework, the "MINT Action Plan 2.0", which successfully integrates federal support with rigorous research and targeted, evidence-based programmes. The German model is particularly instructive in its two-pronged approach: creating a broad, supportive ecosystem for the development of STEM, while simultaneously funding and evaluating specific, scalable interventions.

A comprehensive national strategy: the MINT Action Plan 2.0

The German Federal Ministry of Education and Research (BMBF) spearheads national efforts through its MINT Action Plan 2.0. This strategy is founded upon an understanding that to increase participation in STEM, intervention is required across the entire educational pipeline. It focuses on key fields of action, including strengthening the links between school and out-of-school STEM activities, creating opportunities for parents to support their children's STEM education, and laying the foundations for STEM in primary school.

Evidence-based interventions: the case of online mentoring

Germany's strategic focus is complemented by robust research evaluating specific interventions. One significant example is that of CyberMentor, a large-scale online mentoring programme for secondary school girls (Grades 5–13) in Germany who are interested in STEM. The programme aims to foster this interest, provide career orientation and help girls to build a STEM identity by pairing mentees with female mentors from professional STEM fields.

An empirical evaluation of the programme by Stoeger et al. (2021), analysing nine years of longitudinal data from more than 4,000 participants, provides strong evidence for its effectiveness in achieving these goals. The study found that participation led to significant increases in mentees' self-reported engagement in STEM activities and strengthened their certainty about pursuing a STEM career. While these outcomes are proximal and do not measure final enrolment rates, they are key indicators that this intervention is successful in influencing the pipeline. The study, conducted entirely within Germany, concluded that different mentoring formats (one-on-one, group, hybrid) were all effective, making it a valuable model for scalable interventions. This finding reinforces the general principle of the importance of role models, complementing evidence from various types of interventions, such as the

short, in-person visits studied by Breda et al. (2021), by demonstrating the effectiveness of a long-term, online mentoring approach.

Nuanced insights: the international student population

Research in Germany also provides nuanced insights into specific populations. A study by Preuß et al. (2025) on international university students in Germany found that those in STEM fields, despite being a highly selective group, reported lower academic self-efficacy than international students in non-STEM fields. This highlights that even highly qualified students may struggle with confidence in demanding STEM environments, reinforcing the continued need for post-secondary support (Section 3.3). Interestingly, the study found fewer direct gender differences in expectations and values within this highly self-selected international cohort, suggesting that those who overcome initial barriers to study abroad in STEM may have different motivational profiles.

Key takeaways for the EU from Germany include:

- The value of a holistic national strategy that integrates families, schools and out-of-school partners.
- The proven effectiveness of large-scale online mentoring programmes as a cost-effective and scalable tool for reaching girls across a wide geographical area.
- The importance of continued research into the specific needs of diverse student populations (such as international students) to ensure support measures are effective for all.

Spain

The case of Spain provides an exemplary data-driven diagnosis of the lifecycle of the gender gap in STEM, as documented in a comprehensive 2024 report by EsadeEcPol. This approach – which meticulously traces how early psychological factors cascade into educational and labour market disparities – serves as a model for evidence-based problem identification.

Crucially, the policy solutions proposed are directly tied to this diagnosis and align strongly with the evidence base of effective interventions presented earlier in the present report. The EsadeEcPol report highlights that the gender gap begins early, with girls in Spain exhibiting lower self-perception in mathematics and higher maths anxiety from 4th grade (Cobrerros et al., 2024). This aligns with emerging EU-based longitudinal research suggesting that maths anxiety can be a powerful predictor of choosing a non-STEM educational track (Cuder et al., 2024). Despite showing similar academic performance, girls progressively opt out of maths-intensive pathways, leading to stark underrepresentation in degrees such as Computer Science (13 %) and in vocational training, where only 7 % of female graduates are from STEM-related fields, compared with 52 % of male graduates (Cobrerros et al., 2024).

Diagnosing the gender gap: from early anxiety to career choices

The EsadeEcPol report (Cobrerros et al., 2024) details how the gender gap in STEM is rooted more in psychological and social factors than in innate ability. Key findings include:

- **Educational segregation despite performance:** despite achieving similar or better academic results throughout their education, girls progressively opt out of maths-intensive pathways. In Spain, this is evident in upper-secondary (*Bachillerato*) choices, university entrance exam subjects and, ultimately, university enrolment. The report highlights the stark underrepresentation of women in university degrees such as Computer Science (13 %) and Telecommunications Engineering (23 %), and even more so in vocational training (FP), in which only 7 % of female graduates are in STEM fields, compared with 52 % of male graduates.
- **Labor market disparities:** the gender gap in education translates directly into the workforce. Only 5.5 % of employed women in Spain are in STEM occupations, compared with 13 % of men. Furthermore, the EsadeEcPol analysis reveals a “leaky pipeline”, whereby women with STEM degrees are 2.7 % less likely than their male counterparts to be working in a STEM occupation five years after graduation.

Proposed interventions: addressing stereotypes, pedagogy and support systems

The Spanish report proposes a suite of evidence-based interventions that directly address the factors already identified in the present review, namely:

- **Countering stereotypes and promoting role models:** echoing the causal evidence from Breda et al. (2021) as well as Porter and Serra (2020), the report strongly recommends increasing girls’ access to female role models through mentorships and masterclasses. This is confirmed by a Spanish experimental study on the “Inspiring Girls” programme, which finds that visits by female STEM professionals positively impacted girls’ beliefs in their own success and their desire to choose a STEM career (González-Pérez et al., 2020).
- **Improving pedagogy and curriculum:** the report calls for the curriculum and materials to be free of gender bias, as well as advocating training for teachers in inclusive, collaborative problem-solving methodologies. This is supported by an EU-level analysis (Achiam & Holmegaard, 2016), which advocates for curricula to be reinterpreted through a more socio-scientific and gender-inclusive lens.
- **Building self-efficacy through STEAM:** the EsadeEcPol report (Cobrerros et al., 2024) suggests fostering STEM participation through extracurricular activities that build self-confidence. This approach is corroborated by a systematic review (Bases-García & García-Sánchez, 2021), which concludes that STEAM approaches positively affect self-efficacy – a key target for interventions identified in Section 3.2.

- **Family and systemic support:** the report's proposals include raising awareness among families to increase early maths exposure, as well as supporting policies that ensure equal opportunities in the workplace, such as equitable parental leave and pay transparency, to make STEM careers more attractive and sustainable for women.

Spain's focus on both educational reform and improving labour market conditions for women in STEM offers a holistic policy framework for other EU nations, excellently demonstrating how a data-driven diagnosis can lead to evidence-aligned strategic action.

5. Policy recommendations and future research directions for the EU

The present report has navigated the intricate landscape of STEM education pathways, first by synthesising evidence on the factors that influence students' choices (summarised in Table 2) and then by evaluating the effectiveness of various interventions designed to boost participation (summarised in Table 3). Drawing upon this analysis, this chapter translates key findings into actionable considerations for EU policy development and identifies critical areas in which further research is needed to build a more robust evidence base, particularly within the diverse contexts of the EU. The overarching goal of this is to support the EU and its Member States in effectively achieving its strategic objectives – including the ambitious target of 32 % of student enrolment in higher education being in STEM fields – thereby fostering the larger, more diverse and highly skilled STEM workforce essential to the EU's future prosperity and innovation. This issue is discussed below when we address the core research questions and objectives that have guided this work.

A consolidated overview of the factors that influence STEM choices, discussed in Section 3, can be found in Table 2. Similarly, Table 3 summarises the interventions evaluated in Section 4, including their evidence base and potential cost-effectiveness. These tables provide the synthesised foundation for the recommendations and research directions that follow.

Table 2. Summary of factors influencing STEM choices

Focus Area	Factor	Key finding/impact on STEM choice/participation	Strength of evidence
Early influences	Parental beliefs (utility value)	Beliefs influence student decisions to take courses (causal link shown)	Causal, non-EU
Early influences	Pre-college experiences ("STEM dose")	Advanced experiences correlate strongly with later achievement in STEM	Correlational/suggestive, non-EU
Psychosocial	Self-efficacy/ confidence	Strong correlate/predictor of STEM aspirations/choice	Correlational/suggestive, non-EU
Psychosocial	Motivation/interest (relevance/communal)	Intrinsic interest, perceived relevance, communal framing influence choice	Correlational, EU
Psychosocial	Competitiveness	Predicts choice of maths/science tracks; explains part of gender gap	Causal, EU
Psychosocial	Identity/perceptions (rigidity/belonging)	Negative STEM image deters; sense of belonging is vital for persistence	Suggestive, EU (qual/rev)
Post-secondary	Professor/gender role models	Female role models positively impact female persistence	Causal, EU and non-EU

Focus Area	Factor	Key finding/impact on STEM choice/participation	Strength of evidence
Causal, EU career context	Career priorities (interest/fulfilment)	Intrinsic interest/fulfilment key for current STEM students	Correlational, EU
Career context	Skills/wage dynamics	Rapid skill change	Potential decline in early wage premium observed, US

Notes on evidence strength: “causal” indicates RCTs or strong quasi-experimental (QE) designs; “suggestive” includes weaker QE, strong correlations or qualitative findings. “qual/rev” indicates Qualitative/Review. “Observational/Modelled” relies on observational data/models; “logical/systemic” is based on systems analysis rather than direct intervention studies. “EU/non-EU/US” refers to study location(s).

Table 3. Summary of the effectiveness of interventions that influence STEM choices

Area of focus	Intervention	Key finding/impact on STEM choice/participation	Evidence strength	Potential cost-effectiveness (interventions only)
Targeting individuals	Parental information (utility value)	Increases the taking of courses in STEM (US study)	Causal, non-EU	Low
Targeting individuals	Exposure to role models (peers/professionals)	Increases interest/enrolment in specific fields	Causal, EU	Low–medium
Targeting individuals	Online resilience training (Career WISE)	Increases coping/problem-solving skills (PhD level, US)	Causal, non-EU	Medium
Curriculum/pedagogy	Integrated STEM curriculum	Can improve attitudes/skills (EU examples)	Suggestive, EU	Medium
Curriculum/pedagogy	Framing STEM Communally	Increases interest, especially for women (US studies)	Causal, non-EU	Low
Curriculum/pedagogy	Early digital interventions (confidence)	Can increase girls’ interest/confidence early on (EU study)	Causal, EU	Medium/variable
Curriculum/pedagogy	Creative STEM activities (FabLabs)	Can increase perseverance/creativity (EU study)	Causal, EU	Medium–high
Post-secondary support	Learning communities	Can improve outcomes for women/minorities (US lottery study)	Causal, non-EU	Medium–high
Post-secondary support	Online/blended learning	Potential for equivalent outcomes at lower cost	Causal, non-EU	Potentially lower

Systemic levers				
Higher education	HE funding reform (cost recognition)	Incentive structure significantly impacts institutional STEM supply	Logical/systemic	Policy lever
Educational pathways	Reform of VET-to-HE pathways	Key strategic opportunity to broaden STEM talent pool	Logical/systemic	Policy lever

Notes on evidence strength: “causal” indicates RCTs or strong quasi-experimental (QE) designs; “suggestive” includes weaker QE, strong correlations or qualitative findings. “Observational/modelled” relies on observational data/models; “logical/systemic”, based on systems analysis rather than direct intervention studies. “EU/non-EU” refers to study location(s).

Note on cost-effectiveness: categories are qualitative estimates; robust data are largely absent.

5.1. Factors influencing STEM education choices

The first research question guiding this report explored who chooses STEM and why, as well as who does not, with a focus on differences within the EU. The synthesis of findings confirms that **influencing STEM participation is a complex challenge shaped by a multitude of factors operating at individual, family, educational and broader systemic levels**. Individual aspirations and interests, particularly in mathematics and science, coupled with strong self-efficacy beliefs and academic achievement, emerge as significant drivers of STEM choices. **Support from family, especially parents, as well as exposure to relatable role models, are consistently highlighted as important positive influences. Conversely, negative perceptions of STEM, such as viewing STEM fields as rigid or narrow, or experiencing challenges like maths anxiety, can deter potential students.**

The present analysis highlights that **students’ perceptions are often intertwined with broader societal stereotypes about who “belongs in” and is capable of working in STEM fields**. This merits careful thought with regard to the way in which STEM environments are structured and presented. Findings suggest that **interventions need to actively counter negative stereotypes by emphasising the creative and interdisciplinary aspects of STEM, as well as its societal impact**, thereby broadening its appeal and allowing students to envision constructing desirable identities within these fields. The evidence reviewed also indicates that social factors such as friendship group norms (as discussed in Section 3.2) and the influence of competitiveness play a role in educational sorting, underlining the importance of social context. Furthermore, **previous positive educational experiences** in STEM, particularly during secondary education, are consistently associated with a greater likelihood of pursuing STEM at tertiary level (Schneider & Saw, 2018).

However, from a broader sociological perspective, it is crucial to recognise that although it is necessary to address these individual, familial, and early educational factors, **such measures may be insufficient if they are not accompanied by significant adaptations at system level** (Wang, McGuckin, & Zhang, 2023; McLure & Aldridge, 2022). The effectiveness of interventions that target individual choice and perception is heavily mediated by the structural opportunities, institutional constraints and sociocultural signals embedded within broader educational and societal systems (Bounoua, Sadeh, & Payne, 2024; Weeden, 2002). In the absence of coordinated reforms – such as equitable higher education funding mechanisms aimed at building STEM capacity, and the creation of inclusive pathways from VET – even the most well-designed individual-level programmes may struggle to generate sustainable growth in STEM participation (Molla, 2021; Downes, 2014). This underlines the importance of **systemic alignment**, suggesting that the continued stagnation in STEM enrolment despite policy interventions may stem from the broader institutional context that shapes – and often constrains – the impact of targeted initiatives (Da Wan & Kuzhabekova, 2022).

5.2. Evaluating the effectiveness of interventions

The second research question we have addressed focuses on which measures are effective in helping the EU to achieve its objectives with regard to STEM. This entails evaluating interventions for their effectiveness and considering the EU context. Based on the evidence reviewed, several types of intervention show promise, albeit with varying levels of causal certainty and relevance for direct EU-wide implementation.

Interventions that target individuals and families include parental involvement strategies, such as providing utility-value information about STEM courses. Rigorous US-based evidence shows that **informing parents about the value of maths and science** can causally influence the number of STEM courses their children take. **Role model and mentoring programmes** are also consistently linked to positive outcomes, with strong causal evidence from specific settings that shows exposure to diverse role models can impact persistence and major choice. While these findings suggest potential to increase initial STEM uptake, more rigorous, EU-based causal evaluations of diverse mentoring models are needed. To confirm their effectiveness in various European contexts, it is also necessary to consider how differing higher education funding structures and the associated financial leverage of parents or students might modulate the impact of such supportive interventions.

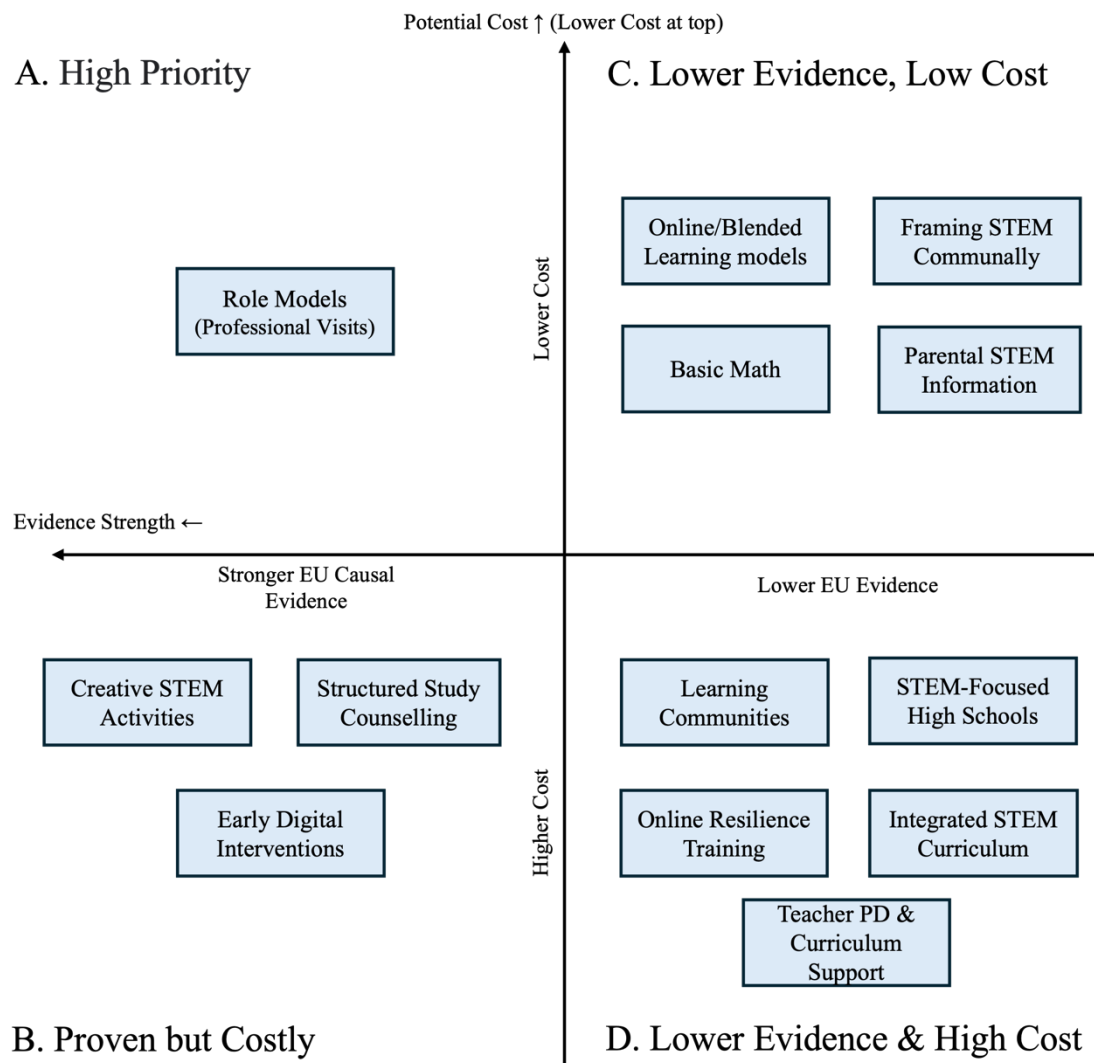
Pedagogical approaches, such as implementing **integrated STEM curricula**, are explored for their potential to make STEM learning more engaging and conceptually coherent. EU-based evidence suggests positive impacts on student attitudes and performance in integrated tasks, implying the potential of such

approaches to foster sustained interest that could lead to increased tertiary enrolment – although further study is required to determine a direct causal link. Efforts to **explicitly address negative perceptions and stereotypes about STEM through curriculum adjustments or communication** are also seen as important, drawing insights from broader bias reduction literature as well as promising early-stage interventions such as digital tools that build confidence in STEM among young girls (demonstrated with causal evidence in an EU context).

Systemic factors represent powerful levers to influence the supply side of the STEM pipeline. Due to complexity and the scarcity of data, there is little direct causal evidence linking specific changes in **higher education funding formulas or VET-to-HE transition policies** to isolated increases in STEM enrolment. However, economic logic and existing analyses strongly suggest such measures have an influence. Funding models that adequately account for the higher costs of quality STEM education can incentivise universities to expand provision, while clear and effective pathways from VET into STEM higher education can significantly broaden the pool of potential entrants.

5.3. Cost-effectiveness considerations

To evaluate interventions, it is also necessary to consider the costs associated with them – an area in which (as this review has found) comprehensive, comparable data are often limited. However, based on the available evidence and the nature of the interventions discussed, some preliminary observations on cost-effectiveness can be drawn. Figure 2 provides a visual summary of the key interventions evaluated in Section 4, mapping them according to two key dimensions. Horizontally, along the x axis, we plot **the strength of the evidence supporting their impact on STEM choice/participation, with stronger evidence on the left and weaker evidence on the right (prioritising causal findings). This reflects their rigour and relevance, and prioritising causal findings**). Vertically, we plot their **estimated potential cost (ranging from low to high, with the lowest at the top of the figure)**. This mapping creates four distinct quadrants (A, B, C, D), offering a strategic overview of the intervention landscape.

Figure 2. Evidence and cost profile of STEM interventions.

Interventions in the “High Priority” quadrant (A) are characterised by strong, EU-based causal evidence and relatively low implementation cost, making them the most compelling candidates for scalable policy action. These interventions are positioned in the upper-left of Figure 2. The primary example fitting this profile is the use of role models, specifically through brief professional visits to classrooms. This approach is supported by a large-scale RCT in France, which demonstrated a significant causal impact on female students’ enrolment in selective STEM programmes (Breda et al., 2021) and appears to be a low-to-medium cost intervention.

Conversely, the “Proven but Costly” quadrant (B) comprises interventions that are also backed by strong EU-based causal evidence, but which require more significant financial or structural investment. These interventions, located in the lower-left, have demonstrated effectiveness but, due to their resource-intensive nature, demand careful cost-benefit analysis and the strategic allocation of resources. Examples that have causal backing from within the EU include hands-on approaches involving creative STEM activities such as those carried out in

FabLabs (Ballerini et al., 2024); personalised support such as structured study counselling (Beckmann & Fervers, 2024); and early digital interventions designed to boost girls' confidence in STEM (Grosch et al., 2022).

The “Lower Evidence, Low Cost” (C) quadrant (upper-right) presents interventions that show great promise for piloting and further evaluation within the EU. These approaches are generally low-cost, but the strongest causal evidence currently supporting them either comes from outside the EU, or existing evidence from the EU is more suggestive. This makes them ideal candidates for testing their transferability. This category includes information-based strategies such as providing parents with information about STEM – a strategy supported by strong non-EU RCT evidence (Harackiewicz et al., 2012). It also includes framing STEM communally in communications, which is backed by non-EU experimental evidence (Diekman et al., 2011). In addition, this quadrant includes digital approaches such as online/blended learning models, the cost-effectiveness of which is supported by a non-EU RCT (Chirikov et al., 2020), as well as targeted support such as basic maths MOOCs, the EU evidence for which is suggestive but not yet strongly causal with regard to pathway choice (Van den Broeck et al., 2020).

Lastly, the “Lower Evidence, High Cost” quadrant (D) contains interventions that represent a higher risk in terms of immediate, large-scale investment, as they combine significant costs with an evidence base that is either primarily non-EU or is less conclusive in establishing impact on initial STEM choice. This (lower-right) quadrant includes effective but costly models with strong non-EU evidence, such as intensive learning communities (Russell, 2017) and STEM-focused high schools (Means et al., 2016). It also includes broad, systemic changes such as comprehensive integrated STEM curricula (Knipprath et al., 2018) and teacher professional development and curriculum support programmes (Lynch et al., 2019), which require more robust evidence establishing a direct causal link to increased higher education enrolment, relative to their high implementation cost.

Systemic interventions that relate to higher education funding and VET pathways are not plotted on the cost/evidence matrix as they represent policy frameworks, involving complex policy changes at national level. However, EU-level activities that supporting the exchange of knowledge and the collection of data about these topics are relatively low in cost but offer potentially high leverage.

Overall, the strategic mapping in Figure 2 helps to guide policy by identifying scalable priorities (Quadrant A), effective but resource-intensive options (Quadrant B), promising areas for piloting (Quadrant C), and interventions where further research is needed to justify significant investment (Quadrant D).

5.4. Evidence-based policy recommendations

To translate the findings above into recommendations, it is necessary to leverage those interventions with the strongest evidence base while also addressing critical systemic factors and acknowledging areas that require further research within the EU. Based on the analysis outlined in the earlier chapters, the EU and its Member States should prioritise the following actions, categorised according to their primary focus:

5.4.1. Recommendations that target individual, family and psychosocial factors

Actively implement and scale up diverse STEM role model initiatives: strong causal evidence demonstrates the positive impact of exposure to relatable role models (including female faculty, successful peers, and professionals) on student choices, persistence and outcomes – especially for underrepresented groups (as discussed in Sections 3.1, 3.3 and 4.1). Therefore, policy makers at all relevant levels should actively design, fund, implement and scale up initiatives that increase the visibility of, and meaningful interaction with, diverse STEM role models throughout secondary and higher education pathways.

Invest in interventions that address perceptions and stereotypes about STEM early and consistently: support the development and scaling of interventions that explicitly aim to counter negative perceptions and stereotypes by emphasising creativity, collaboration and societal impact (see evidence in Sections 3.2 and 4.2). This includes investing in interventions that target younger students with the aim of building confidence and challenging biases early, drawing inspiration from successful EU pilots. Support the development and dissemination of engaging, integrated STEM curricula and pedagogical approaches that reflect this broader image of STEM (discussed in Section 4.2).

Highlight and support cost-effective intervention approaches for individual engagement: encourage Member States to explore and evaluate potentially low-cost, scalable interventions aimed at individuals and their immediate support networks, such as information campaigns targeting parents or students (e.g. regarding the utility of STEM, as detailed in Section 4.1), and well-designed mentoring schemes. Disseminate evidence on the cost-effectiveness of such interventions.

5.4.2. Recommendations that target systemic and institutional factors

Promote policy learning and the exchange of knowledge on systemic reforms: facilitate structured dialogue and peer learning among Member States

on effective strategies for higher education funding models that adequately support high-cost, strategically important STEM programmes (see Section 4.3) and on reforms that create clear, effective pathways from VET into STEM higher education (see Section 4.4). Disseminate best practices and analysis from bodies such as the OECD and EUA with regard to cost-informed funding mechanisms and VET-to-HE transitions.

Strengthen EU-wide data infrastructure for the monitoring and evaluation of systemic performance: as a prerequisite for better understanding STEM pathways, bottlenecks and the impact of policies across diverse contexts, the EU and Member States should invest in developing robust, comparable data infrastructure. This includes supporting initiatives for the longitudinal tracking of student cohorts, detailed information on VET-to-HE transitions (flows and outcomes), and standardised collection of higher education funding and cost data by discipline (the need for which is highlighted throughout Section 1.1 and discussed in relation to funding in Section 4.3).

Pilot and evaluate VET-to-STEM HE bridging mechanisms: acknowledging that VET pathways represent a key strategic opportunity to broaden the STEM talent pool (as highlighted in Section 4.4), dedicate specific funding (e.g. via Erasmus+, ESF+) to pilot diverse VET-to-STEM HE transitions models across interested Member States. These pilots should test concrete mechanisms such as targeted bridging courses, enhanced guidance systems and credit recognition frameworks, incorporating mandatory, rigorous evaluation of their impact on STEM high education enrolment, persistence, and cost-effectiveness (building on the discussion in Section 4.4).

5.5. Key research gaps for the EU

Despite the existing body of literature, significant gaps remain that hinder fully evidence-based policymaking at EU level. A pressing need exists for more EU-based **studies employing rigorous causal designs**, such as RCTs and quasi-experiments, to evaluate the effectiveness of specific interventions aimed at increasing STEM enrolment. Specifically, a key research priority is the systematic and rigorous evaluation of interventions from elsewhere in the world that show causal promise (e.g. parental information campaigns about the utility of STEM, diverse role model initiatives) or those with positive EU-based potential (e.g. integrated STEM curricula, early digital interventions to address confidence) within multiple, diverse EU contexts. Such evaluations are crucial in order to confirm the effectiveness of these interventions, to understand their transferability, and to gather robust data before widespread adoption is considered. This includes generating EU-specific evidence on parental information or engagement strategies, the impact of different models for role model and mentoring programmes, and the effectiveness of interventions designed to counter stereotypes or to broaden perceptions of STEM within

diverse European cultural contexts. The causal impact of integrated versus traditional STEM curricula on long-term higher education choices also requires robust evaluation.

A major gap persists regarding the **cost-effectiveness analysis** of almost all interventions aimed at increasing participation in STEM; future evaluations must systematically incorporate the collection of data on cost and its analysis. Rigorous research is also needed to understand the causal impact of specific features within EU funding models (such as changes in cost weightings or the introduction of performance contracts) on institutional decisions regarding the supply and quality of STEM programmes, and ultimately student enrolment. Similarly, the effectiveness of different VET-to-STEM pathway models in increasing enrolment in STEM higher education requires robust evaluation, including comparative analyses across Member States.

Furthermore, there is a **need for more longitudinal studies** tracking representative cohorts of students through secondary education, vocational training, higher education choices, and into the labour market within diverse EU countries. Such studies are essential to understand the long-term determinants, transitions and outcomes across different educational pathways. Research should also explore how the effectiveness of various factors and interventions varies across the diverse cultural and socioeconomic contexts and education systems of EU Member States and regions. In addition, comparative longitudinal research is needed across EU Member States to understand how career values and priorities related to STEM (including intrinsic motivations, fulfilment and perceptions of communal vs. agentic goals) evolve during adolescence, and how these developmental trajectories influence later STEM choices. Lastly, analysis comparable to studies on skills obsolescence and wage dynamics in other regions is needed using European data in order to understand labour market influences on STEM career choices within the specific context of the EU. Specific research is also required into the factors that influence VET students' aspirations and decisions with regard to transitioning to STEM higher education.

5.6. Methodological recommendations for future research

To effectively address the gaps identified and to build a stronger, EU-specific evidence base for policy, future research into STEM participation should adhere to key methodological principles. Wherever feasible, researchers should employ experimental designs such as randomised controlled trials (RCTs), or quasi-experimental designs that use methods such as regression discontinuity design (RDD), difference-in-differences (DiD), or instrumental variables (IV) approaches, to isolate causal effects and establish a clear link between interventions and outcomes. Research must prioritise studies conducted within representative EU Member States, ideally including multi-country comparative analyses, to capture

and understand the influence of diverse national, cultural, socioeconomic contexts and education systems on the factors and intervention effects. Crucially, EU and national funding mechanisms (e.g. Horizon Europe, Erasmus+, national research programmes) must actively encourage and adequately resource such rigorous evaluations, prioritising studies that incorporate robust causal designs as well as the systematic analysis of cost-effectiveness when examining interventions.

Integrating the collection of cost data and the analysis of cost-effectiveness should become a standard component of all intervention evaluations, in order to provide policymakers with crucial information on the efficiency of different approaches. Investing in and utilising longitudinal datasets that track individuals over extended periods across their educational and career transitions is essential for understanding dynamic influences and long-term impacts. Combining quantitative rigour – necessary for measuring effects and establishing causal links – with qualitative insights, is recommended. Mixed-methods approaches can provide a deeper understanding of the mechanisms behind observed effects, student perceptions and the specific contextual factors that influence choices and the effectiveness of interventions.

Studies should focus on measuring outcomes that are directly relevant to policy goals, such as enrolment in specific STEM fields, successful transition rates from VET into STEM higher education, and metrics related to increasing diversity within STEM pathways. Lastly, fostering closer links and collaboration between researchers and policymakers – ideally from the initial research design phase through to the dissemination of findings – is paramount to ensuring that research questions are policy relevant, findings are understandable and actionable, and the resulting evidence base effectively supports policymaking that aims to increase STEM participation in the EU.

6. Conclusions

Increasing participation in STEM in order to meet the EU's ambitious goals necessitates a sustained, evidence-informed and multi-pronged strategy. This report observes that **the path towards – or away from – STEM careers is not determined by single factors, but rather by a complex interplay of influences** that operate across an individual's lifespan and are embedded within diverse national contexts across the Union.

Our analysis highlights that **early influences, including supportive family environments, parental beliefs about the utility of STEM and engaging in pre-college educational experiences**, lay critical groundwork. As students progress through school, **psychological and social factors become increasingly salient**; their choices are shaped significantly by **strong self-efficacy beliefs, intrinsic motivation and interest** (particularly when this is linked to perceived relevance or communal goals), as well as the development of a compatible STEM identity. Conversely, negative perceptions of STEM fields as being rigid or unwelcoming, as well as the threat of stereotyping and maths anxiety can act as powerful deterrents. The role of competitiveness appears nuanced, influencing educational tracking but potentially being less of a differentiator among those who have already sorted into (or are strongly considering) a STEM trajectory. Furthermore, **post-secondary experiences – in particular, exposure to relatable role models (such as female professors or peers) – as well as supportive learning environments and diagnostic/feedback mechanisms, demonstrably impact persistence and success.**

The evaluation of interventions reveals several promising avenues, although the evidence base varies in terms of its strength or geographical origin. **Interventions for which there is strong causal evidence supporting their positive impacts include providing information to parents about the utility value of STEM** (shown to be effective in the US); **various forms of exposure to role models (female professors, peer/professional visits – with causal evidence from both EU and US contexts); targeted study counselling** (evidence from the EU); **creative STEM activities such as FabLabs (EU evidence); and early digital interventions to boost girls' confidence** (EU evidence). Other approaches, such as integrated STEM curricula, show positive suggestive evidence within the EU. This highlights a crucial point: even the most effective individual-level interventions may see their impact blunted if the overarching systemic constraints remain unaddressed.

Critically, this report underlines that systemic factors are very important and must be addressed simultaneously. Individual choices and the success of targeted interventions are fundamentally shaped by the structure and resourcing of educational systems. Specifically, higher education funding models across the EU must transparently acknowledge and adequately compensate for the

significantly higher costs of delivering quality STEM education, to incentivise sufficient supply and avoid institutional constraints on enrolment. Simultaneously, **creating clearer, more permeable pathways from VET into STEM higher education represents a vital (and potentially under-utilised) strategic opportunity** to broaden the talent pool and enhance social mobility within STEM. Addressing the systemic barriers related to funding and pathways is arguably a prerequisite for achieving substantial increases in STEM participation across the EU.

Despite identifying effective strategies and critical systemic levers, significant gaps persist in the evidence base, hindering fully optimised policymaking. **There is a pressing need for more rigorous, EU-based causal evaluations** of many interventions – in particular, testing the transferability of promising non-EU models (such as parental information campaigns) and assessing diverse role model/mentoring approaches within European contexts. Furthermore, a pervasive lack of comprehensive, comparable data on cost-effectiveness limits the ability to prioritise investments efficiently. Understanding the causal impacts of specific adjustments to funding models and VET pathway reforms requires dedicated research, as does tracking student trajectories via longitudinal studies within diverse EU Member States.

Therefore, **the achievement of the EU's STEM ambitions requires a dual focus. It is crucial to implement targeted, evidence-informed interventions**, such as role model programmes and information campaigns, prioritising those that have already proved effective within the EU, while rigorously testing other promising approaches. However, this report concludes that an exclusive reliance on such initiatives, while valuable, would be insufficient – and furthermore, risks addressing only the symptoms rather than the root causes of constrained STEM participation. The greatest potential for transformative impact at scale is likely to lie in tackling the fundamental systemic barriers related to higher education funding and educational pathways. These “high cost, high benefit” reforms – ensuring funding models can support an expansion in the provision of quality STEM, and creating permeable, well-supported VET-to-HE pathways – are essential prerequisites for sustainable growth. To be successful, EU strategy must therefore strategically combine the implementation of targeted, proven interventions with the political will to enact these foundational systemic reforms. By so implementing this combination, informed by context-specific analysis and a commitment to closing key research gaps, the EU can move decisively closer to building the robust, diverse and highly skilled STEM talent pool that is essential for its future prosperity, innovation and global leadership.

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